THE AGENTIC AI HANDBOOK

Concepts, Design Patterns, and Future Directions

AGENTIC
AI

KAUSHIK BAR

ISBN: 9798265909534

Disclaimer and Legal Notice

Dedication

To my father Mrit, my mother Jas, and my brother Krish,
who have always been constant sources of light throughout my life.

Foreword

"*The Agentic AI Handbook: Concepts, Design Patterns, and Future Directions*" offers a comprehensive exploration of one of the most transformative frontiers in artificial intelligence. Moving beyond traditional reactive and narrow AI systems, agentic AI embodies autonomy, goal-directed reasoning, self-improvement, and the capacity to operate effectively in complex, uncertain environments. This book traces the evolution of agentic systems from conceptual origins in philosophy and cognitive science to cutting-edge technical architectures and real-world applications.

Spanning five major parts, the book begins by establishing the **foundations of agentic AI**, introducing its defining characteristics, conceptual underpinnings, and the core concepts, such as goals, planning, self-reflection, and situatedness, that distinguish agentic systems from conventional AI. It then examines the **technical building blocks** that enable agency: planning algorithms, reinforcement and meta-learning methods, memory and world-modelling systems, reasoning frameworks, and approaches to uncertainty, confidence, and meta-cognition.

Building on these components, the text explores **architectural paradigms**, including Belief–Desire–Intention (BDI) frameworks, cognitive architectures like SOAR and ACT-R, hybrid symbolic and sub-symbolic models, and large language model (LLM)-driven agents. These are followed by discussions on open-ended learning, multi-agent systems, and simulated or embodied agents that bridge virtual and physical environments.

The **applications and impact** section investigates how agentic AI is being deployed in diverse domains such as robotics, autonomous vehicles, scientific research, education, and large-scale automation. It also addresses societal and ethical dimensions, economic disruption, governance, human–machine collaboration, and the moral challenges of deploying autonomous decision-makers.

Finally, the book looks ahead to **future directions**, charting pathways toward Artificial General Intelligence (AGI), exploring principles for safe and interpretable agents, and outlining open research frontiers in emotion-aware, self-repairing systems, and human–AI hybrid cognition. The closing chapter argues that the future of agentic AI will not be dictated solely by technological capability, but by deliberate human choices about how these systems are designed, governed, and integrated into the fabric of society.

With a balance of technical depth and accessible explanation, "*The Agentic AI Handbook*" serves as both a reference for researchers and practitioners, and a guide for policymakers, educators, and curious readers who wish to understand the architectures, possibilities, and responsibilities of building truly intelligent agents. The central thesis is clear: agentic AI should not be built merely to replicate human capacity, but to act as partners in solving the complex, interconnected challenges of the 21st century.

Agentic AI should not seek to replace human capacity, but to augment human potential, serving as trusted partners in creating a safer, more intelligent, and more equitable future.

Overview

"*An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.*" [1].

Artificial Intelligence has made remarkable strides in perception, prediction, and pattern recognition. But these capabilities alone do not make an intelligent agent. Agentic AI represents the next evolutionary leap: systems that can set goals, reason about the world, adapt their strategies, learn continuously, and act autonomously in dynamic environments. Unlike traditional AI that simply reacts to inputs, agentic systems proactively shape their environment in pursuit of objectives, balancing autonomy with human-aligned values.

From autonomous spacecraft exploring distant planets to virtual research assistants accelerating scientific discovery, agentic AI is poised to become a central driver of innovation, productivity, and societal transformation. Yet this leap also brings unprecedented risks, including ethical dilemmas, governance challenges, and safety concerns, that demand careful design and oversight.

Who Should Read This Book

- **AI Researchers & Practitioners** seeking to build or understand autonomous, adaptive systems.
- **Policymakers & Ethicists** designing governance frameworks for autonomous decision-making.

- **Educators & Students** in AI, cognitive science, robotics, and related fields.
- **Industry Leaders** exploring applications in automation, research, and innovation.

<u>Key Takeaways</u>

1. **Agentic AI is the bridge between narrow AI and AGI**, enabling systems that can act with purpose, adapt to change, and collaborate effectively.
2. **Technical and philosophical foundations are equally critical** for building reliable, ethical, and human-aligned agents.
3. **Architectural choices shape capabilities and risks**, from cognitive architectures to LLM-based frameworks.
4. **Applications span every domain** but must be paired with robust safeguards.
5. **The future of agentic AI depends on alignment**, interpretability, and the ability to balance autonomy with human oversight.

Book Structure

This book is organized into **five parts**, each building a comprehensive understanding of agentic AI from philosophical roots to future trajectories.

Part I: Foundations of Agentic AI

We begin by defining agentic AI in contrast to reactive or narrow AI, exploring its origins in philosophy, cognitive science, and robotics. Readers will learn what makes an AI system "agentic": its capacity for goal-directed planning, self-reflection, and context-aware action.

We also explore the philosophical and cognitive dimensions of agency, comparing human and machine intentionality, and introduce key concepts such as situatedness, embodiment, and autonomy.

Part II: Technical Building Blocks

The technical underpinnings of agentic AI are unpacked in detail:

- **Planning and Goal-Directed Behaviour**: From classical planning algorithms to hierarchical task networks.
- **Learning Mechanisms**: Reinforcement learning, meta-learning, and continual learning for adaptation in changing environments.
- **Memory and World Modelling**: Integrating episodic memory with simulated world models to enable foresight and scenario planning.

- **Reasoning and Decision-Making**: Contrasting symbolic reasoning with statistical methods and examining hybrid approaches.
- **Uncertainty and Self-Reflection**: Incorporating probabilistic models, active learning, and meta-cognition to improve reliability and trustworthiness.

Part III: Architectures and Systems

We explore the blueprints for building agentic AI:

- Belief–Desire–Intention (BDI) models
- Cognitive architectures like SOAR and ACT-R
- Hybrid symbolic and sub-symbolic systems
- Large Language Model (LLM)-based agent frameworks

We also examine open-ended learning agents, multi-agent ecosystems, and the design of simulated and embodied agents that learn by doing.

Part IV: Applications and Impact

Agentic AI is already reshaping multiple sectors:

- **Robotics and Autonomous Vehicles**: Decision safety and adaptive navigation.
- **Education**: Personalized tutoring and lifelong learning assistants.
- **Scientific Discovery**: AI-driven hypothesis generation and automated theorem proving.

- **Societal Considerations**: Economic disruption, legal frameworks, political implications, and the future of human–machine collaboration.

Part V: Future Directions

The closing section looks beyond today's capabilities:

- Pathways to Artificial General Intelligence (AGI).
- Design principles for safe and interpretable agents.
- Open challenges in emotion-aware systems, human–AI cognition, and cross-domain intelligence.

The book concludes with a vision for human-cantered agentic AI: systems that are not merely tools, but collaborators in solving the complex, interconnected problems of the future. Code and Bibliography are included in the **Appendix**.

Table of Contents

PART I: Foundations of Agentic AI

Artificial intelligence (AI) is entering a new phase of development that is marked not only by reactive and pattern-based behaviour but by systems capable of initiating action, pursuing goals, and adapting autonomously. This marks the emergence of what many scholars and technologists are beginning to call Agentic AI. **Part I** lays the theoretical foundation for understanding this paradigm shift, exploring the conceptual underpinnings, historical emergence, and the distinctions that separate agentic systems from earlier generations of AI.

By exploring the origins, philosophical grounding, and essential definitions of agentic AI, it equips the reader with a precise mental model of what it means for an AI to act with agency. The principles and distinctions introduced here are not purely theoretical; they provide the interpretive lens through which the technical designs in **Part II**, the architectural frameworks in **Part III**, and the real-world applications in **Part IV** should be understood. Without this grounding, discussions about algorithms, decision-making strategies, or ethical safeguards risk being decoupled from the very essence of agency. In this way, **Part I** acts as the conceptual bedrock for both the practical engineering and the societal considerations presented in later parts of the book, ensuring that all discussions remain anchored in a coherent and well-defined vision of agentic intelligence.

Agentic AI represents a turning point: systems that are no longer just tools executing commands but entities with internal models, plans, preferences, and the ability to act autonomously within bounded environments. As AI transitions from automation to agency, a host of philosophical, technological, and ethical questions surface. What does it mean for a machine to have a goal? Can AI be truly autonomous without consciousness? How do we measure responsibility or intent in such systems? The answers to these questions begin with a careful dissection of what agentic AI is, and what it is not.

Chapter 1. Introduction to Agentic AI

As artificial intelligence continues to evolve, a fundamental shift is taking place. This change transcends beyond predictive analytics, pattern recognition, and static task execution. Instead of viewing AI solely as a tool that reacts to stimuli, a new paradigm is emerging. AI as an agent can formulate goals, plan actions, interact proactively with its environment, and even reflect on its performance. This chapter introduces the concept of agentic AI, outlining its significance, origin, and defining attributes.

The term *agentic* derives from the word "agency," a concept deeply rooted in both cognitive science and philosophy, which denotes the capacity of an entity to act independently and make choices. When applied to machines, agency implies a level of initiative, autonomy, and sustained action that exceeds reactive or command-based models. Agentic AI systems are those that do not merely respond to prompts. Rather, they initiate, adapt, and persist in pursuit of goals.

Characteristic	Traditional AI	Autonomous Systems	Agentic AI
Adaptability	Limited; relies on pre-programmed rules or retraining on fixed datasets.	Moderate; adapts within predefined operational parameters.	High; dynamically learns and adapts to new goals, environments, and contexts without external reprogramming.

Decision-Making Autonomy	Low; requires frequent human oversight for novel situations.	Moderate; capable of operating independently in familiar domains.	High; can independently set sub-goals, reprioritize tasks, and modify strategies in response to environmental changes.
Self-Reflection Capabilities	None; no capacity to evaluate its own reasoning or performance.	Minimal; may perform system health checks but lacks meta-cognition.	Present; incorporates meta-cognitive processes for assessing decision quality, uncertainty, and potential improvement pathways.

Table 1.1: Comparative Characteristics of Agentic AI, Traditional AI, and Autonomous Systems.

With the rise of foundation models, reinforcement learning agents, and open-ended AI architectures, the notion of agentic intelligence is no longer speculative. We are beginning to see AI systems capable of task decomposition, iterative planning, memory retention, and even reflective reasoning. Some of these characteristics were once considered unique to human cognition. Yet, this evolution raises foundational questions:

- What is agentic AI, and how is it different from traditional AI systems?
- What theoretical foundations support its development?
- What capabilities define an agentic system?

- Why does this matter for the future of science, business, and society?

These questions form the basis of this chapter. We begin by defining agentic AI in conceptual terms, then trace its historical emergence, and finally explore the key features that distinguish it from more conventional forms of artificial intelligence.

What is Agentic AI?

Agentic AI refers to artificial intelligence systems that are capable of autonomous goal-directed behaviour. Unlike conventional AI models that passively respond to inputs based on learned patterns, agentic AI systems actively initiate actions, plan trajectories toward objectives, monitor progress, and adapt strategies based on feedback from the environment.

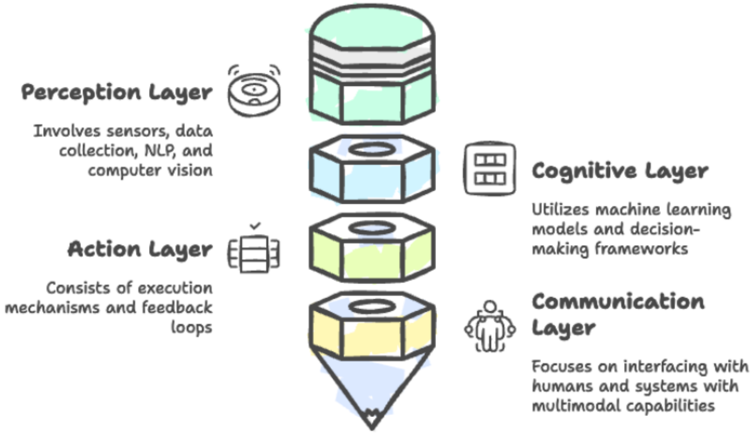

Figure 1.1: Components of agentic AI (source: Arion Research).

Agentic AI is not merely a theoretical concept. It is increasingly being used across domains:

- **Autonomous Research Agents:** Tools like AutoGPT or SciFact Agents conduct literature reviews, run simulations, and write reports autonomously.
- **Game AI:** AlphaStar and OpenAI Five exhibit sustained, strategic planning.
- **Digital Personal Assistants:** ReAct or Voyager-style agents that browse the web, schedule appointments, and book travel with minimal input.
- **Robotic Process Automation (RPA):** Agentic bots in finance or HR that initiate, complete, and review end-to-end workflows.
- **Military and Space Exploration:** NASA's autonomous rovers operate with significant agentic capabilities.

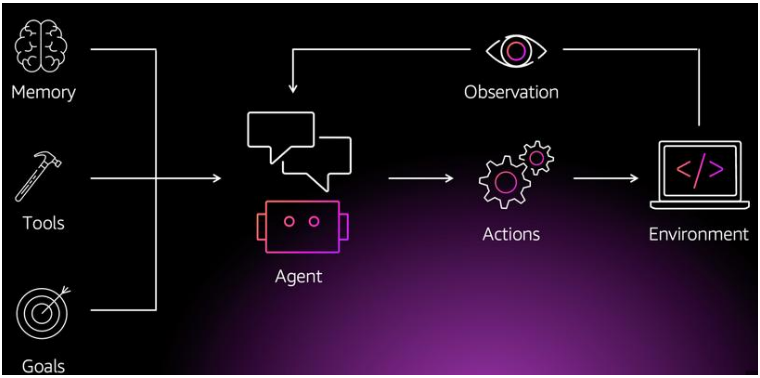

Figure 1.2: A basic agentic AI pipeline (source: AWS).

In simpler terms, an agentic AI is not just *smart*; it is *motivated*. It operates based on an internal representation of goals and engages with the world in pursuit of these goals, modifying its actions and beliefs as circumstances evolve.

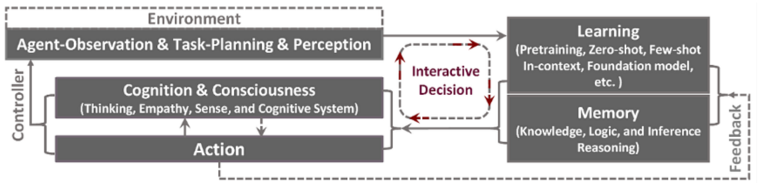

Figure 1.3: An agentic AI paradigm for supporting embodied multi-modal generalist agent systems [2].

This agentic property can be formalized in terms of four key attributes:

Attribute	Description
Autonomy	The ability to act without external control or instruction
Intentionality	Goal-oriented behaviour that reflects internal states or preferences
Adaptivity	Continuous learning and adjustment in response to feedback or environment
Persistence	Sustained action over time directed toward objectives

Table 1.2: Summary of agentic properties / attributes.

Agentic AI builds upon this foundational view but goes beyond it. Whereas traditional agents follow fixed policies or behaviours derived from a training set, agentic AIs are characterized by their ability to restructure their policies, question prior knowledge, and pursue long-term objectives, often without direct human oversight.

Figure 1.4: Positioning Agentic AI in the AI Spectrum.

The following code snippet introduces a minimal rule-based autonomous agent for a hands-on experience.

```
python

# Minimal Agentic AI Example - rule-based navigation
class SimpleAgent:
    def __init__(self):
        self.state = "idle"

    def perceive(self, environment):
        if environment.get("obstacle"):
            return "avoid"
        return "move_forward"

    def act(self, perception):
        if perception == "avoid":
            self.state = "turn_left"
        else:
            self.state = "move_forward"

agent = SimpleAgent()
env = {"obstacle": False}
perception = agent.perceive(env)
agent.act(perception)
print(f"Agent state: {agent.state}")
```

Origins and evolution

The development of agentic AI can be traced through three intertwined trajectories in AI history:

1. **Agent-based Systems in Early AI (1960s–1980s):** Early AI research was dominated by symbolic reasoning and rule-based systems. Projects like SHRDLU [3] introduced the idea of AI systems interacting with the world using internal models. These systems had some notion of goals but lacked real autonomy or adaptability.
2. **DARPA Autonomous Land Vehicle (ALV) Program (1980s-1990s):** The ALV program, initiated by the U.S. Defence Advanced Research Projects Agency, was one of the earliest large-scale attempts to build a vehicle capable of navigating real-world terrain without constant human control. Equipped with early computer vision systems, lidar, and sensor fusion techniques, the ALV demonstrated that AI could integrate perception, planning, and control in a continuous loop, an early form of situated autonomy. While limited in adaptability, it laid the groundwork for modern autonomous systems that must reason about changing environments.
3. **Soar Cognitive Architecture (1990s)**: The Soar architecture, developed by John Laird, Paul Rosenbloom, and Allen Newell, became a cornerstone in cognitive AI research. Unlike specialized expert systems, Soar aimed to model *general cognitive processes*, integrating decision-making, learning, and problem-solving into a unified framework. Over the years, enhancements such as reinforcement learning integration and episodic memory modules pushed Soar closer to agentic behaviour by allowing agents to learn from prior experiences and adapt their strategies over time.
4. **Reinforcement Learning and Rational Agents (1990s–2010s):** The 2000s saw significant breakthroughs in embodied AI: robots with physical presence interacting

in real-world settings. Projects like Honda's ASIMO and Boston Dynamics' early quadrupeds demonstrated the importance of embodiment in adaptive agents. Advances in tactile sensing, real-time control, and mobility not only improved physical autonomy but also encouraged research into *cognitive embodiment*: the idea that intelligence emerges partly from physical interaction with the environment. With the rise of reinforcement learning (RL), AI began to be conceptualized as agents interacting with environments. RL introduced the concept of reward-maximization as a computational proxy for agency [4], but most RL agents still lacked long-term planning and reflection.

5. **Emergence of Agentic Architectures (2015–Present):** By the late 2000s, research began to emphasize systems that blended statistical machine learning with symbolic planning. Hybrid architectures allowed robots and software agents to simultaneously *perceive* their environment, *reason* about possible actions, and *adapt* when unexpected changes occurred, which are core traits of agentic AI. Recent advances in large-scale models, multi-agent systems, and embodied AI have enabled the emergence of truly agentic systems. Tools like OpenAI's AutoGPT or Google's PaLM-E represent the first generation of AIs capable of initiating tasks, searching for information, and even reasoning about their limitations [5].

These milestones collectively represent a shift from isolated, task-specific AI systems toward integrated, self-directed agents: a trajectory that has accelerated in the 2010s and 2020s with advances in large-scale neural architectures, multimodal perception, and autonomous decision-making frameworks.

Milestone Year	Breakthrough	Contribution to Agentic AI
1972	SHRDLU	Symbolic reasoning in structured worlds
1998	Sutton & Barto's RL	Learning from interaction
2016	AlphaGo defeats Lee Sedol	Goal-driven AI under uncertainty
2021	Gato by DeepMind	Generalist agent with multiple modalities
2023	AutoGPT / BabyAGI / Voyager	Self-improving, task-initiating agents

Table 1.3: Summary of agentic AI evolution.

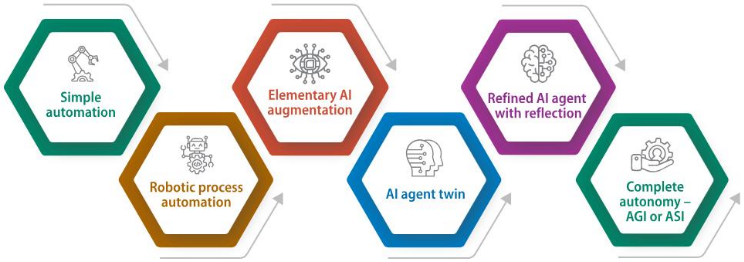

Figure 1.5: Progress in the field of agentic AI (source: Infosys).

How is it different from traditional AI?

Traditional AI systems, even the most powerful large language models (LLMs), do not qualify as agentic unless they meet specific criteria. Below is a comparison to elucidate the differences:

Feature	Traditional AI	Agentic AI

Input Handling	Passive	Active seeking of goals/tasks
Goal Representation	Implicit (e.g., next token)	Explicit, modifiable objectives
Autonomy	Limited (responds to prompts)	High (can initiate actions unprompted)
Learning Mechanism	Static post-training	Online, self-improving
Planning and Reflection	Minimal	Multi-step reasoning with self-evaluation
Embodiment	Often disembodied	Often embedded in virtual/physical environments
Persistence Over Time	Session-bound	Maintains state across tasks

Table 1.4: Distinguishing features of agentic AI over traditional AI.

To illustrate the difference, consider this comparison:

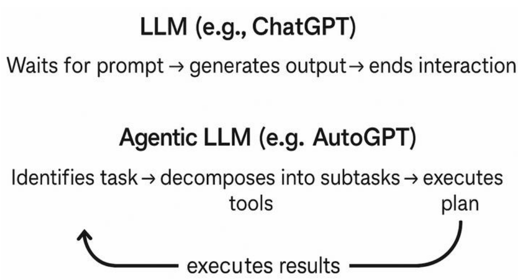

Figure 1.6: Traditional vs. Agentic AI Pipeline.

To build agentic AI systems, several architectural components are required beyond those used in typical deep learning pipelines:

1. **Goal Management System:** Stores and updates goals; includes mechanisms for prioritization and goal decomposition.

2. **Planning Engine:** Generates action sequences from goals using reasoning or search methods.
3. **Memory Module:** Maintains working and long-term memory to track past experiences and inform future decisions.
4. **Perception-Action Loop:** Interfaces with the environment to observe consequences and take actions.
5. **Self-Reflection Mechanism:** Evaluates its own decisions, identifies failures, and updates internal models.

Component	Description
Goal Manager	Maintains and updates objectives
World Model	Internal representation of external environment
Action Selector	Chooses next action based on plans and goals
Feedback Integrator	Uses environment signals to update beliefs and behaviours
Reflective Core	Monitors self-performance and generates meta-goals

Table 1.5: Functional Components of an Agentic System.

The cognitive components in an agentic system are:

Cognitive Function	System Component	Implementation Example
Memory	Episodic + Semantic	Vector DBs, transformer memory
Planning	Goal → Action Steps	PDDL, STRIPS, language-based planning
Action	Environment interfacing	APIs, robotics modules
Reflection	Meta-cognition	Prompt-chaining, confidence scoring

Table 1.6: Cognitive Blueprint of an Agentic System.

The next chapter discusses the philosophical and cognitive underpinnings of agentic AI.

Chapter 2: Philosophical and Cognitive Underpinnings

The concept of *agency* is deeply rooted in philosophy, cognitive science, and psychology. Before computational systems could approximate intelligence, the question of what it means to act with intention and autonomy preoccupied thinkers for centuries. As artificial intelligence (AI) systems evolve beyond reactive automation into decision-making entities capable of self-directed behaviour, the relevance of such philosophical and cognitive constructs intensifies.

This chapter explores the foundational differences between human and machine agency, then examines key notions in both conceptual and computational contexts. These are not just abstract concepts; they have direct implications for how agentic AI should be conceptualized, designed, and governed.

The three key concepts pertaining to the philosophical and cognitive underpinnings of agentic AI are:

- Consciousness
- Intentionality
- Autonomy

Consciousness

Consciousness is one of the most debated concepts in philosophy and cognitive science. From a human perspective, it encompasses awareness, experience, and self-reflection. Philosophers like Thomas Nagel have famously asked, "*What is it like to be a bat?*", emphasizing the subjective quality of conscious experience [11].

AI systems, however, do not have qualia (the raw feel of experience). They can simulate aspects of consciousness, such as attention or working memory, but lack phenomenal consciousness [12]. While some argue that sufficiently complex architectures may eventually support some form of machine consciousness [13], this remains speculative.

Attribute	Humans	AI Systems
Phenomenal Consciousness	Present	Absent
Access Consciousness	Present	Simulated (via attention models)
Self-awareness	Developed (mirror test, etc.)	Absent or rudimentary

Table 2.1: Comparison of Consciousness in Humans vs. AI.

Intentionality

Intentionality, in philosophy, refers to the "aboutness" of mental states. Our thoughts are about something [14]. Humans think about goals, consequences, and abstract concepts. Machines, by contrast, do not *intend* in this way. Their behaviour may be goal-directed, but it stems from algorithmic design, not mental representation.

In agentic AI, intentionality is simulated through planning modules, reinforcement learning objectives, or utility functions. For example, a robot vacuum cleaner may avoid obstacles and return to its charging station, behaviours that seem intentional, yet arise from simple rules and reward optimization.

Autonomy

Autonomy in AI refers to the degree to which a system can make decisions independently of direct human control. Operational autonomy allows systems to act within constraints, like drones following GPS paths or financial bots trading based on thresholds.

However, ethical autonomy, the ability to evaluate consequences and make value-based decisions, is currently beyond AI capabilities. Some researchers are exploring value alignment and machine ethics [15], aiming to align AI systems with human values, but this remains a nascent field.

Level	Human Analogy	AI Example
Reactive Autonomy	Reflex actions	Rule-based agents
Tactical Autonomy	Short-term goal adjustment	RL agents in video games
Strategic Autonomy	Long-term planning, goal setting	Multi-agent systems with planning
Ethical Autonomy	Moral deliberation	Not yet achieved in AI

Table 2.2: Levels of Autonomy.

Agency in humans vs. machines

Human agency

Human agency is often defined as the capacity to act intentionally and to make choices based on reasoning, emotion, and a sense of self [6]. In philosophy, particularly in the tradition of existentialism and phenomenology, agency is closely linked with free will, responsibility, and moral reasoning [7].

Dimension	Human Agency
Consciousness	Subjective awareness of thoughts and sensations
Intentionality	Goal-directed mental states
Autonomy	Self-governance, informed by values and context
Moral Reasoning	Capacity for ethical evaluation of actions
Reflexivity	Ability to think about one's own thoughts
Embodiment	Tied to the biological and physical body
Situatedness	Influenced by historical, cultural, and social factors

Table 2.3: Summary of human agencies.

Humans possess first-person perspective, an ability to reflect on their own beliefs, desires, and decisions. This reflexivity (i.e., the capacity to be aware of one's own mental states) is a crucial differentiator. Also, agency in humans is inherently embodied and situated. That is, our actions are always situated within a cultural, physical, and social environment that shapes our intentions and constraints [8].

Machine agency

Machine agency, in contrast, refers to the capacity of AI systems to perform tasks, make decisions, or adapt behaviour in ways that *appear* autonomous or intentional, but are fundamentally grounded in programmed objectives, learning algorithms, and input-output mappings [9]. Unlike human agents, machines lack a subjective internal perspective, that is, they do not *feel* or *understand* their actions.

That said, with the emergence of agentic AI systems that operate semi-autonomously—such as intelligent assistants, robotic agents, or multi-agent systems—we begin to ascribe forms of *limited agency* to machines. However, this agency is typically instrumental, designed to serve human goals rather than emerge intrinsically.

Dimension	Machine Agency
Consciousness	Absent; no subjective awareness
Intentionality	Emergent via goals encoded in algorithms
Autonomy	Limited to operational parameters
Moral Reasoning	None, unless explicitly programmed heuristics
Reflexivity	Simulated via meta-learning or self-monitoring
Embodiment	Optional (e.g., robots vs. software agents)
Situatedness	Limited unless context-aware models are implemented

Table 2.4: Summary of machine agencies.

This contrast reveals that machine agency is a metaphor, grounded in functionality rather than experience. Yet, as machines exhibit increasingly complex behaviour, from navigating traffic to negotiating contracts to managing information, the perception of agency becomes more socially and economically significant [10].

The following code snippet shows a simple machine agent that evaluates options based on utility vs. human-style preference weighting.

```
python

# Utility-based decision
options = {"OptionA": 0.8, "OptionB": 0.5, "OptionC":
0.3}
decision_machine = max(options, key=options.get)
print("Machine chooses:", decision_machine)

# Human-style weighted choice (introducing
bias/emotion factor)
human_bias = {"OptionA": -0.1, "OptionB": 0.3,
"OptionC": 0.0}
weighted_options = {k: options[k] + human_bias[k] for
k in options}
decision_human = max(weighted_options,
key=weighted_options.get)
print("Human chooses:", decision_human)
```

Types of AI agents

Simple reflex agents

Simple reflex agents operate on fixed condition–action rules, reacting directly to environmental triggers without referencing past events. These systems are fast but rigid, offering predetermined responses to specific inputs. **Examples**:

- Keyword-based spam filters
- Preprogrammed chatbots
- Automated email responders

Model-based reflex agents

Model-based reflex agents maintain an internal model built from past observations, allowing them to infer and respond to aspects of the environment they can't currently observe. While they still use condition–action rules, this added context enables more flexible decision-making. **Examples**:

- Smart thermostats
- Robotic vacuum cleaners with mapping capabilities
- Sensor-driven irrigation systems

Goal-based agents

Goal-based agents plan actions by assessing possible future states and selecting those that lead toward defined objectives. They rely on goal-oriented algorithms to handle complex decision-making tasks. **Examples**:

- AI-powered chess programs
- Route optimization tools
- Customer service chatbots with problem-solving abilities

Utility-based agents

Utility-based agents use utility functions to weigh and compare potential outcomes, aiming for the highest overall benefit. They are designed to make optimal choices in situations involving uncertainty, trade-offs, or competing objectives. **Examples**:

- Self-driving vehicle systems
- Investment portfolio optimizers
- AI-assisted medical diagnosis tool

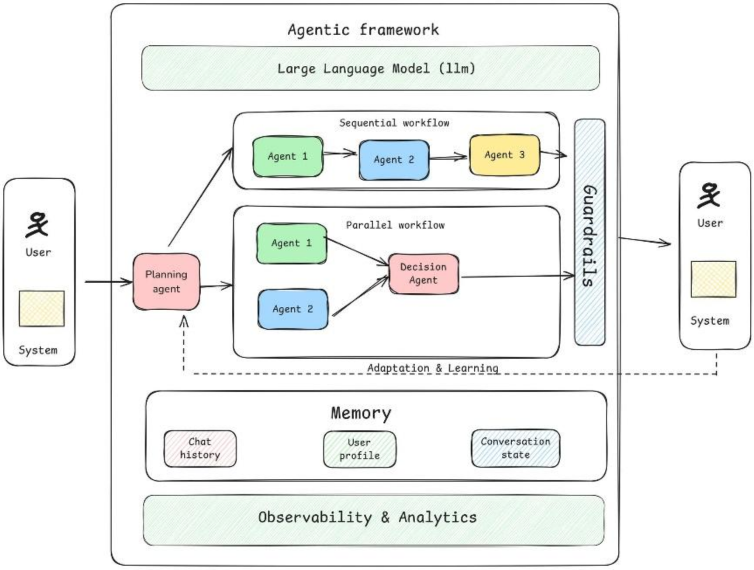

Figure 2.1: Agentic AI architecture (source: Swapan Rajdev, Haptik).

Key takeaways

The exploration in this chapter reveals that **agentic AI remains far from achieving human-level agency**, particularly in aspects tied to subjective experience and moral reasoning. Nevertheless, as AI systems become more adaptive, goal-driven, and context-aware, the need to design and evaluate them through an agency-informed lens becomes increasingly urgent.

In the next chapter, we will dissect the core constructs of agentic AI, such as goal formation, planning, self-monitoring, and embodiment, drawing connections between philosophical notions and computational frameworks.

Chapter 3: Core Concepts and Definitions

The concept of agency, i.e., the capacity to initiate, sustain, and reflect upon goal-directed behaviour, is at the heart of both cognitive science and artificial intelligence. In traditional AI, systems were largely reactive or narrowly task specific. In contrast, agentic AI introduces the potential for autonomy, long-term planning, adaptability, and self-monitoring, i.e., features that more closely resemble how humans operate in dynamic, uncertain environments.

This chapter explores the foundational concepts that define agentic AI systems: goal formation, planning, self-reflection, situatedness, and embodiment. These constructs do not just describe what an agent can do, rather they establish the structural and cognitive scaffolding necessary for robust autonomous behaviour. As AI continues to evolve toward more general forms of intelligence, understanding and formalizing these dimensions becomes essential.

Goals

Goals are representations of desired future states that guide agent behaviour. In humans, goal formation emerges from internal motivations, social cues, and contextual stimuli. In agentic AI systems, goals are typically:

- **Explicitly defined** by designers (e.g., maximize reward, minimize cost).
- **Inferred** through interaction or inverse reinforcement learning [16].
- **Learned adaptively** through environment exploration and internal value systems.

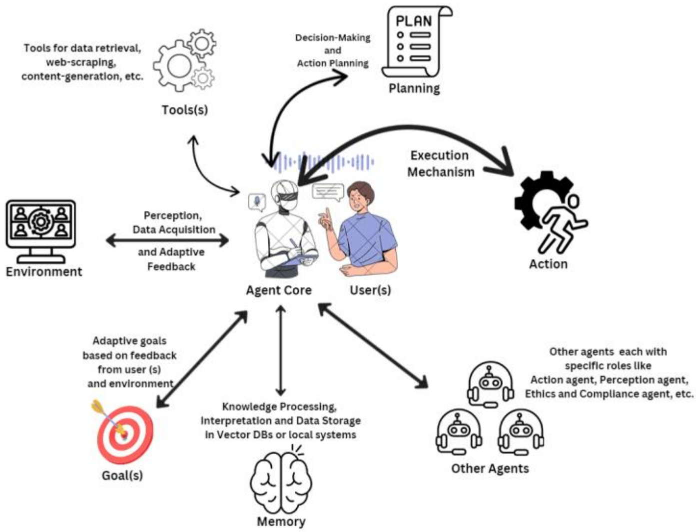

Figure 3.1: Goal-directed behaviour of agents (source: Anthropic).

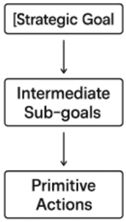

Figure 3.2: Hierarchical Goal Decomposition.

A goal is not merely a target. It plays a regulatory role by shaping action selection, resource allocation, and temporal structuring of behaviour.

Type	Example

Instrumental	Navigate to location X
Strategic	Maximize long-term resource utilization
Social	Coordinate with peers in a multi-agent system
Intrinsic	Minimize prediction error or maximize curiosity

Table 3.1: Types of Goals in Agentic AI.

Agentic AI frameworks often employ hierarchical goal models, where high-level strategic objectives decompose into sub-goals, which in turn translate to low-level actions [17].

Planning

Planning is the process through which agents determine a sequence of actions to move from an initial state to a goal state. Planning in AI involves both search (to explore possible action sequences) and reasoning (to evaluate their utility).

There are three main classes of planning relevant to agentic AI:

- **Classical planning** (deterministic, fully observable domains).
- **Probabilistic planning** (stochastic environments with uncertainty).
- **Hierarchical planning** (multi-level abstraction of tasks and strategies).

Agentic AI systems often use planning approaches such as Markov Decision Processes (MDPs), partially observable MDPs (POMDPs), or Monte Carlo Tree Search (MCTS) to simulate action outcomes before execution ([18]; [19]).

Feature	Relevance to Agentic AI
Goal-directedness	Plans emerge in response to active goals

Flexibility	Plans adapt to environmental changes
Temporal abstraction	Multi-step foresight is crucial for autonomy
Integration with memory	Past experience informs plan generation

Table 3.3: Properties of Agentic Planning.

Self-Reflection

Self-reflection, i.e., the ability to evaluate one's own behaviour, goals, and decisions, is a hallmark of human intelligence and a key enabler of adaptive agency. In AI, self-reflection is implemented through mechanisms like:

- **Meta-learning**: Learning how to learn [20].
- **Model introspection**: Evaluating the uncertainty or correctness of predictions.
- **Policy adjustment**: Updating decision strategies based on performance.

Self-reflective AI agents incorporate **meta-cognition**, that is, they can reason not only about the world but about their own knowledge, limitations, and performance. This is vital for safety, robustness, and continual learning.

Example: A self-driving car detects degradation in sensor reliability during a storm and shifts from high-speed autonomous driving to a cautious, assisted-driving mode. This adjustment involves self-reflective awareness of internal system states and external risks.

Situatedness

Situatedness refers to an agent's ability to operate meaningfully within a specific environment, informed by real-time context, history, and sensorimotor feedback [21]. Agentic AI must go beyond abstract reasoning and act *within* physical or digital worlds where uncertainty, partial observability, and noise are the norm. The main characteristics of Situated AI are:

- **Context-awareness**: Recognizing situational factors (e.g., location, time, risk),
- **Interaction history**: Using past experience to guide present decisions,
- **Environment coupling**: Dynamically adapting behaviour to external feedback.

For instance, a home assistant robot that learns the user's daily routine, adjusts lighting, suggests activities, and navigates changing physical layouts is demonstrating situated behaviour. This real-time loop is critical for dynamic environments such as robotics, conversational agents, and autonomous vehicles.

Embodiment

Embodiment is the grounding of cognition in a physical (or simulated) body that interacts with the environment. Embodiment influences:

- Sensorimotor coordination
- Spatial reasoning
- Emotional and affective behaviour

According to the embodied cognition hypothesis [22], intelligence arises not just from computation, but from how an organism (or AI agent) experiences the world through its body.

In robotics, embodiment is literal: an AI agent has a body with sensors and actuators. In virtual environments (e.g., simulation games, digital assistants), embodiment can be partially realized through digital avatars or conversational interfaces.

Dimension	Physical Robots	Virtual Agents
Sensorimotor loops	Rich and complex	Simulated or abstracted
Environmental coupling	Strong (real-world physics)	Controlled or predefined
Risk and failure	High (e.g., falling, breaking)	Low (simulation restarts)
Social presence	Physical proximity	Expressed via voice or visuals

Table 3.4: Comparison Table: Physical vs. Virtual Embodiment.

Embodiment has implications for trust, human-AI interaction, and learning. For instance, children interacting with embodied robots often show higher engagement and empathy than when using tablet-based agents [23].

Key takeaways

This chapter establishes the functional and cognitive infrastructure necessary for agentic AI. While early AI focused on symbolic reasoning or statistical learning, agentic AI integrates goals, plans, context, and self-awareness, a richer model that mimics some of the complexities of human action. In the next part of the book, we will explore how these foundations are implemented in practice across different architectures and real-world applications.

Concept	Description	Relevance to Agentic AI
Goals	Internal representations of desired states	Anchor planning and drive behaviour
Planning	Sequencing of actions toward goal fulfilment	Enables foresight and adaptability
Self-reflection	Evaluation of internal models, decisions, and performance	Enhances robustness, learning, and safety
Situatedness	Operating meaningfully within an environment	Critical for real-time interaction and learning
Embodiment	Grounding cognition in a physical or simulated body	Shapes perception, action, and social understanding

Table 3.5: Summary of Core Concepts.

PART II: Technical Building Blocks

While the foundations of agentic AI are rooted in philosophical inquiry and conceptual understanding, building truly agentic systems demands a deep integration of technical components that enable autonomy, adaptability, and reflective intelligence. **Part II** delves into the computational and architectural underpinnings that bring agentic intelligence to life.

Agentic AI systems must go beyond pattern recognition and prediction. They must perceive, plan, act, learn, and self-correct in open, dynamic environments. This requires assembling a coherent technical stack that includes:

- Perceptual systems capable of interpreting multimodal input.
- Memory and knowledge representations that support episodic and semantic recall.
- Planning mechanisms that allow long-term goal fulfilment and flexible decision-making.
- Learning algorithms that adapt across contexts and tasks.
- Meta-cognitive modules that monitor internal states and outcomes.

What differentiates agentic AI from conventional AI is not any single component, but how these components interact in a closed-loop system capable of self-guided behaviour and continual improvement. Agentic AI systems must operate under uncertainty, reason about partial information, recover from failure, and re-align with long-term goals, all while interacting with humans and complex environments.

Throughout this section, we will explore how concepts such as deliberative planning, reactive control, reinforcement learning, self-supervised learning, episodic memory, and situated reasoning are combined into cohesive agentic architectures. We will also examine how modularity, communication, and hierarchical control help scale agentic capabilities across diverse domains, from robotics and software agents to virtual companions and decision support systems.

By the end of this part, readers will gain a comprehensive understanding of how the theoretical ideals of agentic AI map onto tangible engineering designs, and what it takes to build systems that not only act but understand why they act.

Chapter 4: Planning and Goal-Directed Behaviour

Planning is a defining characteristic of agency. It transforms passive systems into autonomous agents capable of goal-directed behaviour over time. In both human and artificial cognition, planning involves selecting and organizing actions to achieve desired future states from a current state. For agentic AI, planning bridges perception and action, giving rise to intentional agency that can pursue long-term objectives in complex environments.

Planning is not merely execution of learned behaviours. Instead, it involves symbolic reasoning, temporal foresight, and often contingency management. For agentic systems, planning becomes particularly important when:

- Goals are abstract, multi-step, or delayed in reward.
- The environment is partially observable or dynamic.
- The agent has limited resources and must optimize its actions.
- Goals change over time and require re-planning or goal arbitration.

This chapter explores the computational frameworks that enable such behaviours, starting from the formalism of classical planning and moving towards more flexible and human-aligned approaches like Hierarchical Task Networks (HTNs).

The following table compares the approaches.

Aspect	**Classical Planning**	**HTN Planning**

Goal Specification	Logical formula	Task networks
Plan Structure	Flat sequences	Hierarchical (nested)
Reusability	Limited	High (via methods)
Human Interpretability	Low	High
Scalability	Moderate	Higher due to decomposition
Planning Strategy	State-centric	Task-centric

Table 4.1: Classical Planning vs HTNs.

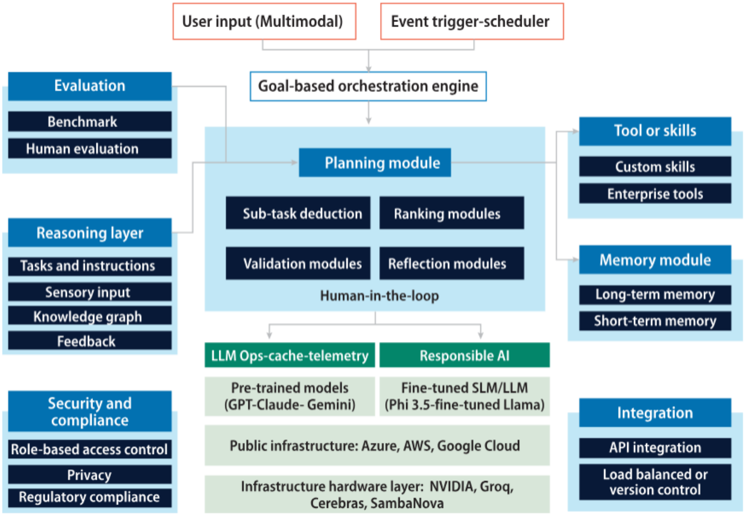

Figure 4.1: Agentic AI planning framework (source: Infosys).

<u>Classical planning</u>

What is classical planning?

Classical planning refers to a well-established branch of AI that models the agent's behaviour as a sequence of actions transforming an initial state into a goal state. It assumes:

- A fully observable, deterministic, and static world.

- A known set of actions with preconditions and effects.
- A clear goal state defined by logical conditions.

Despite these assumptions, classical planning provides a rigorous foundation for structured reasoning and is used in robotics, logistics, and game AI.

Formal definition

A classical planning problem is defined as a tuple $P = \langle S, A, T, s_0, G \rangle$ where:

- S: Set of all possible world states
- A: Set of available actions
- T: Transition function $T: S \times A \rightarrow S$
- s_0: Initial state
- G: Set of goal states (defined via logical predicates)

Each action $a \in A$ is defined as: $a = \langle pre(a), eff(a) \rangle$ where:

- $pre(a)$: Preconditions for applying the action
- $eff(a)$: Effects of the action (state transformation)

The solution to a classical planning problem is a sequence of actions: $\pi = \langle a_1, a_2, \dots, a_n \rangle$ such that: $T(T(\dots T(s_0, a_1,), a_2), \dots, a_n) \in G$.

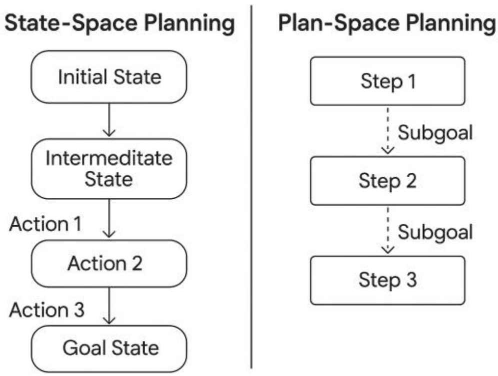

Figure 4.2: Difference between state-space planning and plan-space planning.

Planning languages

The most widely used language for representing classical planning problems is the Planning Domain Definition Language (PDDL). PDDL separates the *domain* (actions, predicates) from the *problem* (initial state, goal). **Example** of PDDL:

```
lisp

(:action move
    :parameters (?from ?to)
    :precondition (and (at ?from) (connected ?from
?to))
    :effect (and (not (at ?from)) (at ?to)))
```

This describes a `move` action that can be taken if you're at a location and it is connected to another. After the `move`, you're at the new location.

The following code snippet provides a simple state-space planning example using BFS.

```
python
```

```
from collections import deque

def bfs(start, goal, neighbors):
    visited = set([start])
    queue = deque([(start, [start])])
    while queue:
        node, path = queue.popleft()
        if node == goal:
            return path
        for neighbor in neighbors(node):
            if neighbor not in visited:
                visited.add(neighbor)
                queue.append((neighbor, path +
[neighbor]))
    return None

neighbors = lambda n: [n+1, n-1] if 0 <= n <= 5 else
[]
path = bfs(0, 5, neighbors)
print("Plan:", path)
```

Planning algorithms

Classical planning algorithms include:

- **Forward Search** (Progression Planning): Explores paths from the initial state towards the goal.
- **Backward Search** (Regression Planning): Works backward from the goal to find a plan from the initial state.
- **Heuristic Search** (e.g., A*): Uses heuristics like the number of unsatisfied goals to guide search.
- **SAT-Based Planning**: Encodes planning as a Boolean satisfiability (SAT) problem.

Limitations of classical planning

While classical planning provides clear semantics and efficient algorithms, it struggles in real-world settings due to:

- Rigid assumptions (full observability, determinism),
- Poor scalability to continuous or stochastic environments,

- Lack of goal abstraction or task hierarchy.

These limitations motivate richer models like Hierarchical Task Networks (HTNs).

Hierarchical Task Networks (HTN)

What is HTN?

HTNs extend classical planning by allowing task decomposition. Instead of focusing solely on atomic actions, HTNs define high-level tasks that can be broken down into subtasks, mirroring how humans plan.

This leads to more flexible, scalable, and semantically rich plans, especially useful for agentic AI that must plan at multiple levels of abstraction.

Formal definition

An HTN problem is defined as: $P = \langle S, T, M, s_0, t_0 \rangle$ where:

- S: Set of world states
- T: Set of tasks (primitive and compound)
- M: Methods for decomposing compound tasks
- s_0: Initial state
- t_0: Initial task network (can include multiple tasks)

There can be following types of tasks:

- Primitive Tasks: Correspond to executable actions (e.g., "move", "pick-up").
- Compound Tasks: High-level objectives (e.g., "clean house") that require decomposition.

HTN Methods

HTN Methods specify how to break a compound task into subtasks. For example:

```
lisp

(:method clean-house
    :task clean-house
    :subtasks (vacuum-floor, dust-furniture, take-
out-trash))
```

Each method may have preconditions and ordering constraints.

The following code snippet provides a minimal HTN example for making tea.

```
python

# Hierarchical Task Network
def make_tea():
    return boil_water() + brew_tea() + serve()

def boil_water():
    return ["Fill kettle", "Turn on stove"]

def brew_tea():
    return ["Place teabag in cup", "Pour hot water"]

def serve():
    return ["Add sugar/milk", "Serve tea"]

print("HTN Plan:", make_tea())
```

Benefits with HTNs

HTNs align well with agentic intelligence due to their:

- **Hierarchical structure**: Supports abstraction, which is essential for long-term planning.

- **Modularity**: Easy to integrate with symbolic or learned components.
- **Reusability**: Methods can be applied across contexts.
- **Human-alignment**: Similar to how people break down goals.

The diagram below illustrates a compound task (Prepare Dinner) decomposed into subtasks through methods.

```
Goal: Prepare Dinner
 ├── Cook Pasta
 │    ├── Boil Water
 │    ├── Add Pasta
 │    └── Drain
 └── Make Salad
      ├── Wash Veggies
      ├── Chop
      └── Mix
```

Figure 4.3: HTN Decomposition Example.

Planning with HTNs

Planning in HTNs involves:

1. **Task selection**: Choose a compound task to decompose.
2. **Method selection**: Select a suitable method for the task.
3. **Recursive decomposition**: Until only primitive tasks remain.
4. **Plan validation**: Ensure that preconditions are satisfied in the current state.

HTN planners include:

- **SHOP2** (Simple Hierarchical Ordered Planner)

- **PyHOP** (Python-based HTN planner)
- **Hierarchical Reinforcement Learning (HRL)** frameworks in learning-based settings

Key takeaways

For agentic AI, the ability to set, pursue, and revise goals in a structured and hierarchical way is essential. HTNs provide an elegant framework for this. Moreover, combining HTNs with learning enables agentic systems to acquire new plans from data or demonstrations (e.g., Inverse HTN Learning).

Modern systems may integrate:

- Symbolic HTN planning for high-level reasoning,
- Neural modules for perceptual grounding,
- RL for low-level control.

Thus, HTNs serve as a bridge between classical symbolic reasoning and modern learning-based AI, which is a hallmark of agentic systems.

Chapter 5: Learning in Agentic Systems

At the core of agentic AI lies the **capacity to learn**, i.e., to not only react to an environment but also adapt to it, reflect on its own experience, and improve over time. Learning distinguishes truly agentic systems from hard-coded automation. An agent capable of learning can evolve its internal models, modify its behaviour, and pursue new goals in unfamiliar environments.

Agentic learning involves more than just fitting models to data. It implies learning through interaction, adapting to task distributions, and retaining knowledge over time without catastrophic forgetting. In this chapter, we explore three key paradigms that enable such capabilities in agentic systems:

- **Reinforcement Learning** for learning through interaction and feedback,
- **Meta-learning**: for learning how to learn across tasks,
- **Continual Learning**: for preserving and updating knowledge over time.

Together, these approaches contribute to the design of adaptive, autonomous agents that improve with experience, adjust to novelty, and operate reliably over extended lifespans.

Reinforcement learning

Reinforcement Learning (RL) enables agents to learn optimal actions through trial and error in an environment. Unlike supervised learning, which learns from labelled datasets, RL relies on reward signals to guide behaviour.

In the context of agentic AI, RL forms the foundation for goal-seeking, self-improving, and autonomously evolving agents.

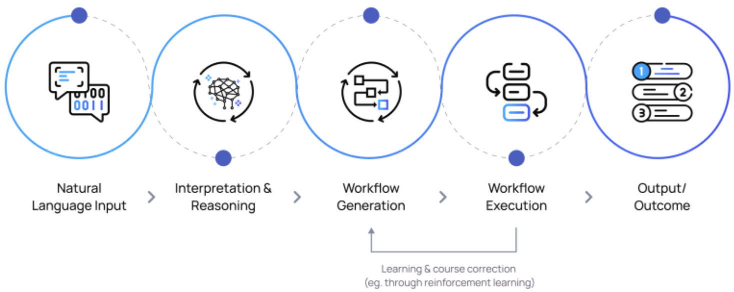

Figure 4.1: RL in agentic AI (source: Aisera).

In 2020, a leading e-commerce company deployed a fleet of mobile robots trained using deep reinforcement learning (DRL) to optimize warehouse item-picking and delivery routes. Traditional rule-based path planning often struggled to adapt to dynamic changes in aisle congestion, battery levels, and shifting priorities.

By framing warehouse navigation as a Markov Decision Process (MDP), the DRL agents learned policies that balanced speed, energy efficiency, and collision avoidance. Over time, the agents demonstrated emergent behaviours such as pre-emptively avoiding traffic bottlenecks and anticipating high-demand areas based on historical patterns.

This deployment bridged the gap between theory and application in agentic AI:

- **Adaptability**: Agents continuously refined strategies through real-time learning from environmental feedback.

- **Decision Autonomy**: No human intervention was required for micro-level route adjustments.
- **Self-Improvement**: Performance improved over weeks of operation without retraining from scratch.

This example illustrates how RL's principles (reward maximization, exploration, and policy optimization) enable autonomous systems to thrive in complex, dynamic environments.

Formulation

RL problems are typically modelled as Markov Decision Processes (MDPs) defined by the tuple:

$$\langle S, A, T, R, \gamma \rangle$$

- S: Set of states
- A: Set of actions
- $T(s'|s,a)$: Transition probabilities
- $R(s,a)$: Reward function
- γ: Discount factor for future rewards

The agent's goal is to learn a **policy** $\mu(a|s)$ that maximizes the expected cumulative reward:

$$\mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t)\right]$$

The following agentic features get enabled by RL:

- **Autonomy**: RL allows agents to operate without explicit programming.
- **Exploration**: Agents actively explore unknown parts of their environment.

- **Delayed gratification**: RL learns to optimize long-term outcomes, not just immediate results.

The following code snippet provides a Q-learning snippet for a grid world.

```
python

import numpy as np
import random

states = range(6)
actions = [0, 1]  # 0: left, 1: right
Q = np.zeros((len(states), len(actions)))
alpha, gamma, epsilon = 0.1, 0.9, 0.1

def step(state, action):
    if action == 1:
        return min(state + 1, 5), 1 if state == 4
else 0
    else:
        return max(state - 1, 0), 0

for episode in range(500):
    state = 0
    while state != 5:
        if random.random() < epsilon:
            action = random.choice(actions)
        else:
            action = np.argmax(Q[state])
        next_state, reward = step(state, action)
        Q[state, action] += alpha * (reward + gamma *
np.max(Q[next_state]) - Q[state, action])
        state = next_state

print("Q-table:\n", Q)
```

Deep Reinforcement Learning (DRL)

Modern RL systems use deep neural networks to approximate policies or value functions. This has led to major breakthroughs in domains such as:

- Game playing (e.g., AlphaGo, OpenAI Five)

- Robotics (e.g., locomotion, manipulation)
- Recommendation systems (e.g., Netflix, YouTube)

Agentic AI benefits from model-based RL as well, where agents learn a model of the environment to plan ahead, more consistent with the planning capacities discussed in the previous chapter.

Limitations

While powerful, standard RL suffers from:

- Sample inefficiency (needs millions of interactions),
- Lack of transferability across tasks,
- Poor adaptation in non-stationary environments.

These challenges motivate the need for meta-learning and continual learning, discussed next.

Meta-learning

Agentic systems are expected to generalize quickly to new tasks based on limited experience, akin to how humans adapt to new challenges using prior knowledge. Meta-learning, also known as “learning to learn,” addresses this need.

In meta-learning, the agent learns across a distribution of tasks, developing an ability to rapidly adapt to new, unseen tasks.

Meta-learning enables agentic systems to:

- Generalize across environments,
- Adapt policies for different goals,
- Learn new skills from few demonstrations.

For example, a home assistant could use meta-learning to personalize itself to different households with minimal data.

Formulation

Meta-learning problems are defined over a task distribution $p(T)$. For each task T_i, the agent observes a support set (training examples) and is evaluated on a query set.

The agent optimizes for fast adaptation and generalization.

The following code snippet demonstrates meta-learning loop pseudo-code.

```
python

import numpy as np

def meta_learning(tasks, meta_lr=0.01):
    meta_params = np.random.rand()
    for task in tasks:
        # Adapt parameters for this task
        adapted_params = meta_params - meta_lr *
np.gradient(task_loss(meta_params))
        meta_params += meta_lr * (adapted_params -
meta_params)
    return meta_params

def task_loss(params):
    return (params - 0.5)**2  # Dummy loss

tasks = [task_loss for _ in range(5)]
final_params = meta_learning(tasks)
print("Meta-learned params:", final_params)
```

Approaches

1. **Gradient-Based Meta-Learning** or **Model-Agnostic Meta-Learning (MAML)**

   - Learns a shared initialization from which gradient descent quickly adapts to new tasks.
   - Examples: Reptile, First-Order MAML (FOMAML).
2. **Metric-Based Meta-Learning**
   - Learns distance functions in embedding space.
   - Examples: Matching Networks, Prototypical Networks.
3. **Memory-Based Meta-Learning**
   - Uses external memory to retrieve prior task experiences.
   - Examples: MetaNet, SNAIL.

Challenges

- Computational complexity,
- Sensitivity to task distribution,
- Lack of interpretability in learned meta-parameters.

Despite these, meta-learning remains a key enabler for flexible, general-purpose agentic intelligence.

Continual learning

Agentic AI systems operate over long timescales and dynamic environments. They must retain previous knowledge while acquiring new capabilities, a problem known as continual learning or lifelong learning.

Continual learning enables agents to:

- Adapt over long-term deployments.
- Maintain a coherent memory of past experiences.
- Avoid regressions in previously learned capabilities.

These are essential for robust and stable behaviour in evolving real-world environments.

Humans learn throughout life, combining **episodic memory** (specific experiences) with **semantic memory** (general knowledge). Agentic AI aspires to mirror this dual learning process for more human-aligned intelligence.

Formulation

In continual learning, an agent faces a sequence of tasks $T_1, T_2, \dots, T_n$ and must:

- Learn each task without forgetting (**catastrophic forgetting**) prior tasks.
- Reuse prior knowledge to accelerate learning (**positive transfer**).
- Store knowledge efficiently.

Approaches

1. **Regularization-Based**: Penalizes changes to parameters important to past tasks. Example: Elastic Weight Consolidation (EWC).
2. **Replay-Based**: Stores a small buffer of past examples. Examples: Experience Replay, Generative Replay.
3. **Parameter Isolation**: Allocates different subnetworks for different tasks. Examples: Progressive Networks, PackNet.
4. **Modular Architectures**: Use dynamic modules that specialize per task.

Key takeaways

While each of the paradigms explored in this chapter is powerful individually, true agentic learning may require their integration. For example:

- Meta-RL combines meta-learning with reinforcement learning
- Continual Meta-Learning allows adaptation without forgetting
- Hierarchical RL with memory supports planning across timescales

Learning Paradigm	Objective	Strengths	Limitations
Reinforcement Learning	Learn from interaction to maximize rewards	Autonomy, exploration, goal alignment	Sample inefficiency, slow adaptation
Meta-Learning	Learn to learn across tasks	Fast adaptation, generalization	Complex training, task sensitivity
Continual Learning	Learn over time without forgetting	Long-term adaptation, memory retention	Task interference, scalability

Table 5.1: Summary of different learning paradigms.

In the following chapters, we explore how these learning foundations integrate with perception, memory, and self-reflection, further pushing toward fully autonomous agentic systems.

Chapter 6: Memory, Perception, and World Modelling

At the heart of any agentic system lies the ability to perceive, remember, and reason about the world it operates in. While learning helps an agent acquire new knowledge, its capacity to act intentionally over time depends on its internal representations of experience and the external environment. These representations are constructed and updated through **perception**, organized and retained through **memory**, and used to form **world models**, enabling anticipation, planning, and adaptation.

This chapter explores three tightly linked components of agentic intelligence:

- **Perception**: how agents interpret raw sensory data,
- **Memory**: how agents encode and recall relevant past experiences,
- **World Modelling**: how agents build internal representations of external reality.

We emphasize the importance of episodic memory for learning from specific past events and explore the use of simulated environments for grounding and testing perception and world models.

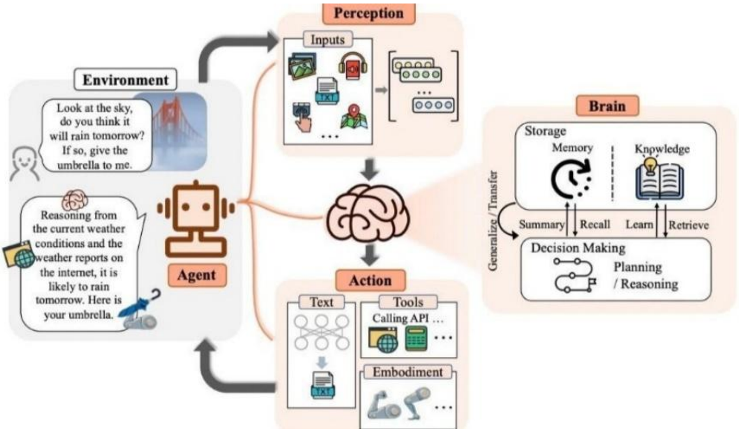

Figure 6.1: Agentic AI and its ability to perceive, remember, and reason (source: LinkedIn).

Below is a conceptual diagram representing the interaction between perception, memory, and world modelling in agentic AI:

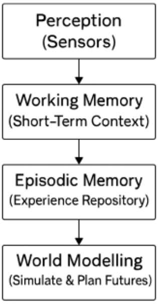

Figure 6.2: Interaction between perception, memory, and world modelling.

Perception

Perception refers to the process of acquiring, processing, and interpreting sensory information. In agentic AI, perception is not passive. It enables decision-making and must be selective, goal-directed, and context-aware.

Sources of perception may include:

- Visual inputs (images, video)
- Audio signals (speech, environmental sounds)
- Tactile signals (pressure, temperature)
- Proprioception (internal sensing of joint/motor states)

In artificial agents, these are often represented as **multi-modal** data streams processed via sensors and neural models.

Active perception

Unlike traditional systems that consume data passively, agentic systems engage in active perception, choosing *what* to perceive and *when* based on task demands. For example:

- A robotic agent may move its camera to inspect an object.
- A software agent may query an API only when needed to reduce bandwidth or latency.

This attention-based filtering mimics human perception, where awareness is contextually modulated.

Learning from perception

Modern systems rely on deep learning models (e.g., CNNs, Transformers) for visual and auditory perception. However, agentic AI extends this with:

- **Perception for action**: Filtering observations relevant to a task.
- **Perception for learning**: Using observations to refine models or detect novelty.
- **Uncertainty-aware perception**: Recognizing when perception is unreliable or incomplete.

Memory

Memory allows agentic systems to retain context, learn from experience, and reason about past interactions. Without memory, every situation would be interpreted anew, rendering learning and intentionality nearly impossible.

Agentic AI systems require at least three forms of memory:

Type	Description
Working Memory	Short-term storage for reasoning or holding temporary beliefs
Episodic Memory	Long-term storage of specific experiences
Semantic Memory	Long-term structured knowledge (e.g., facts, rules)

Table 6.1: Types of memory in agentic AI.

Memory augmentation

Modern agentic systems often rely on memory-augmented neural networks (MANNs), such as:

- Neural Turing Machines (NTM)
- Differentiable Neural Computers (DNC)
- Episodic Memory Networks (EMNets)

These architectures enhance reasoning, adaptation, and few-shot learning, aligning with goals of agentic adaptability.

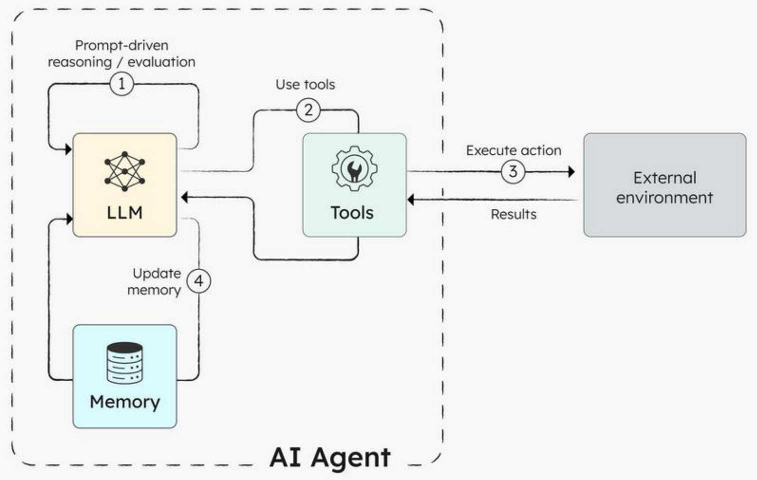

Figure 6.3: Agentic learning with the help of memory (source: Orkes).

Learning from memory

Episodic memory refers to the explicit recollection of specific past events, including context, time, and outcomes. It is critical for:

- Learning from experience,
- Identifying cause-effect relationships,
- Predicting outcomes based on prior scenarios.

In humans, episodic memory is associated with the hippocampus and prefrontal cortex, enabling recall of *what*, *where*, and *when*.

Episodic memory in AI can take many forms:

- Key-value memory banks.
- Recurrent neural networks with long-term dependencies.

- External memory modules (e.g., Differentiable Neural Computers).

The following code snippet demonstrates a simple episodic memory store and retrieval.

```
python

class EpisodicMemory:
    def __init__(self):
        self.memory = []

    def store(self, episode):
        self.memory.append(episode)

    def retrieve(self, condition):
        return [ep for ep in self.memory if
condition(ep)]

memory = EpisodicMemory()
memory.store({"state": "hungry", "action": "eat"})
memory.store({"state": "thirsty", "action": "drink"})

print(memory.retrieve(lambda e: e["state"] ==
"hungry"))
```

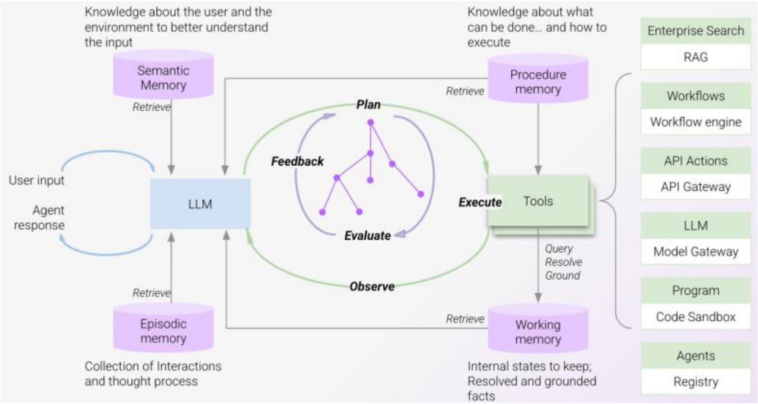

Figure 6.4: Different types of memory used in agentic AI (source: Moveworks).

These memories are queried to recall prior states, actions taken, and rewards received, especially useful for reinforcement learning and planning.

World modelling

A world model is an internal, structured representation of the environment that allows an agent to simulate outcomes, plan trajectories, and generalize knowledge. It serves as the agent's mental map of how the world works.

Agentic systems benefit from model-based reasoning, predicting and simulating future events without acting.

Following are the benefits of world models for Agentic AI:

- Enables counterfactual reasoning (e.g., "What if I had chosen another action?").
- Facilitates fast adaptation via simulation.
- Reduces need for real-world trial-and-error.
- Supports intrinsic motivation (e.g., curiosity).

Key components

Key components of a world model are:

- **State representation**: What is the current state of the world?
- **Transition dynamics**: What happens when an action is taken?
- **Reward model**: What is the expected outcome?

The table below summarizes the types of world models in agentic AI.

Model Type	Description
Model-free	No explicit world model (e.g., standard RL)
Model-based	Learns state transitions and rewards explicitly
Hybrid models	Combine model-free and model-based components

Table 6.2: Types of world models.

Learning world models

World models can be learned using:

- **Supervised learning**: Predict next state given current state and action.
- **Unsupervised learning**: Learn latent dynamics (e.g., via VAEs, predictive coding).
- **Reinforcement learning**: Optimize models to maximize predictive utility.

Important architectures include:

- World Models [24]
- Dreamer [25]
- PlaNet [26]

Many of these models allow agents to *dream* or *imagine* futures, enabling **imagination-based planning**.

The following code snippet simulates two possible future worlds to pick the better one.

```
python

def simulate_world(state, action):
    if action == "gather_food":
```

```
        return state + 5
    elif action == "explore":
        return state + 2
    return state

current_state = 10
actions = ["gather_food", "explore"]
futures = {a: simulate_world(current_state, a) for a
in actions}
best_action = max(futures, key=futures.get)
print("Chosen action based on simulation:",
best_action)
```

Simulated environments

Simulation provides:

- Safe and fast environments to train agentic systems
- Controlled settings for evaluation
- Mechanisms for domain randomization, improving generalization

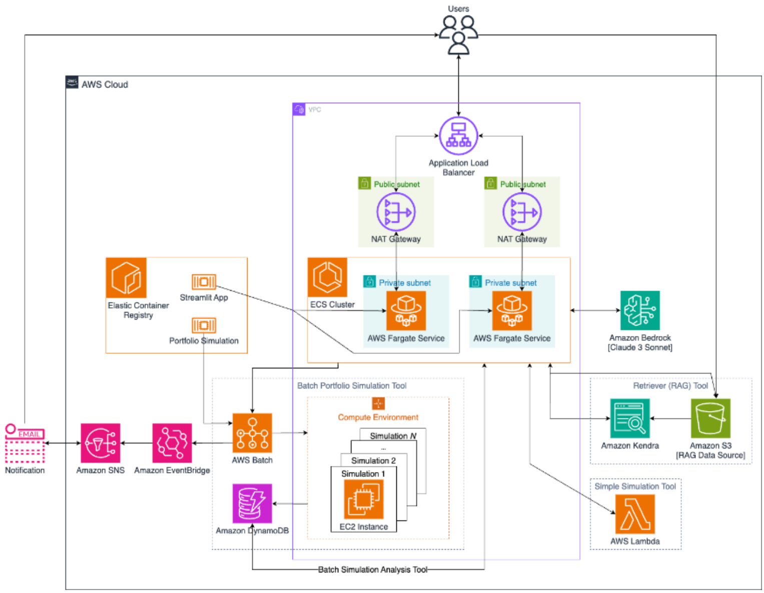

Figure 6.5: An AI simulation assistant with agentic workflows (source: AWS).

Simulated environments offer scalability, reproducibility, and rich interaction. All of these are critical for training and benchmarking agentic systems. Following are some notable platforms for simulated environments:

Environment	Domain	Features
OpenAI Gym	General RL	Modular API, broad task variety
DeepMind Lab	3D navigation & vision	Visual realism, memory testing
Meta-World	Robotic control	50+ robot manipulation tasks
Crafter	Minecraft-style survival	Testbed for exploration, memory, goal pursuit

Procgen	Procedurally generated games	Generalization benchmarks

Table 6.3: Platforms for agentic AI simulated environments.

Simulated success does not always translate to real-world competence due to:

- Perception mismatch
- Action transfer difficulties
- Overfitting to simulation dynamics

Agentic AI must therefore incorporate domain adaptation, sensorimotor grounding, and real-world fine-tuning.

Key takeaways

- **Perception** allows an agent to filter, focus, and interact meaningfully with its environment.
- **Memory**, especially episodic, enables learning from prior experiences and prevents information loss.
- **World models** are critical for simulation, planning, and decision-making.
- **Simulated environments** provide scalable testbeds but require careful alignment with real-world complexity.

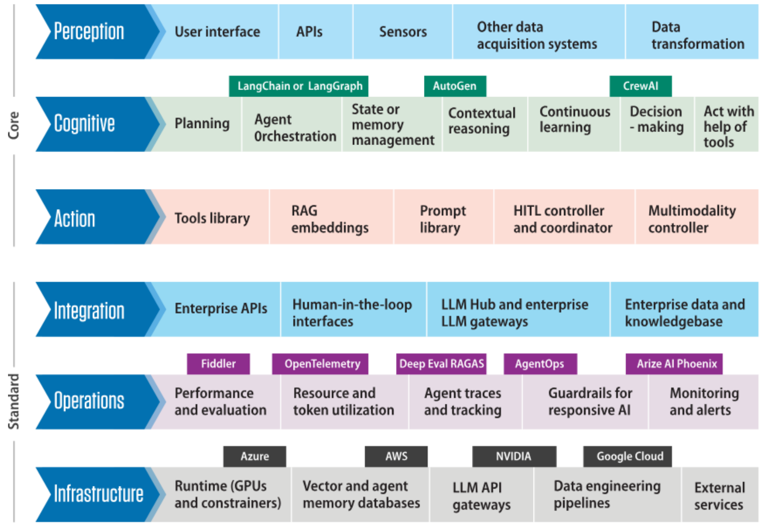

Figure 6.6: Agentic AI layers (source: Infosys).

Chapter 7: Reasoning and Decision-Making

Reasoning and decision-making are central faculties in intelligent systems, whether biological or artificial. In the context of agentic AI, these processes not only enable agents to draw conclusions, make plans, and select optimal actions but also form the cognitive bridge between perception, memory, and behaviour. Unlike passive or purely reactive systems, agentic systems must reason under uncertainty, deliberate about goals, and decide based on both internal models and external feedback.

Consider an assistive robot in a home environment tasked with fetching objects and responding to verbal instructions. A purely symbolic system may understand the logic behind object relationships (e.g., "a mug is likely to be in the kitchen"), while a statistical system could learn from visual data where mugs are most often found.

A hybrid system might:

1. Use deep learning to process visual input (object detection).
2. Use symbolic logic to infer relationships between tasks and locations.
3. Use probabilistic reasoning to update beliefs about object locations.
4. Plan actions using a hierarchical task planner.
5. Adapt future behaviour based on feedback.

Such integration is crucial for developing agentic systems that are both intelligent and explainable.

In this chapter, we explore the evolution of reasoning paradigms in AI, from symbolic logic to statistical inference, and examine how hybrid architectures are emerging as powerful frameworks for enabling agency. The chapter begins by contrasting symbolic and statistical reasoning approaches, discussing their respective strengths and limitations. It then explores how hybrid systems aim to combine the interpretability and generality of symbolic reasoning with the flexibility and scalability of data-driven statistical models.

Symbolic reasoning

Symbolic reasoning refers to the use of formal logic systems to represent and manipulate knowledge through symbols, rules, and logical inference. This approach dominated the early decades of artificial intelligence, especially during the era of "Good Old-Fashioned AI" (GOFAI), where the mind was seen as a symbol-manipulating system [27].

Advantages:

- Transparency and Interpretability: Every decision can be traced through explicit logical rules.
- Transferability: Rules can be applied across domains if the symbols are aligned.
- Modularity: Knowledge can be structured hierarchically.

Disadvantages:

- Brittleness: Symbolic systems often fail when encountering noise or incomplete data.
- Manual Encoding: Requires hand-crafted knowledge bases.
- Scalability: Difficulty dealing with large or high-dimensional input spaces.

Knowledge representation and logic

In symbolic systems, knowledge is typically encoded using:

- First-order logic (FOL) or predicate logic
- Semantic networks
- Ontologies (structured representations of concepts and relationships)

Agentic systems can use these tools to reason about goals, constraints, and the causal structure of the environment. For example, a planning agent might use the STRIPS representation to define preconditions and effects of actions in a symbolic space [28].

Statistical and probabilistic reasoning

With the rise of machine learning, particularly in the 1990s and 2000s, AI systems began leveraging probabilistic methods and data-driven models for reasoning and decision-making.

Advantages:

- Handles noise and uncertainty well.
- Learns patterns directly from data.
- Scales to high-dimensional problems.

Disadvantages:

- Lack of interpretability.
- Requires large amounts of data.

- Struggles with abstract reasoning and commonsense knowledge.

Learning

Statistical reasoning is often embedded within machine learning algorithms. These methods allow agents to learn policies or value functions from experience, a hallmark of adaptive behaviour. Examples:

- Decision trees and random forests
- Probabilistic graphical models
- Deep learning models with Bayesian interpretations

Statistical reasoning allows agents to make inferences under uncertainty. Common models include:

- **Bayesian Networks**: Encode conditional dependencies between variables [29].
- **Markov Decision Processes (MDPs)**: Used for sequential decision-making under uncertainty.
- **Hidden Markov Models (HMMs)**: For temporal pattern recognition.

Agentic AI systems can use such models to infer hidden causes, update beliefs based on evidence, and estimate the likelihood of outcomes.

Hybrid cognitive architectures

To realize truly agentic behaviour, AI systems must combine the strengths of both symbolic and statistical reasoning. This is especially crucial for tasks that require both flexibility and understanding (e.g., a domestic robot deciding how to assist a person based on spoken instructions, past experience, and a semantic understanding of household tasks).

Neuro-symbolic AI

Neuro-symbolic systems aim to integrate neural networks (for perception and pattern recognition) with symbolic reasoning (for logic and planning). Examples include:

- **Differentiable logic layers**: Where logical operations are embedded in neural architectures.
- **Semantic parsing models**: That map natural language to logical forms.
- **Neuro-symbolic concept learners**: That can generalize from few examples [30].

These architectures are promising for creating agents that can both learn from data and explain their reasoning.

Cognitive architectures

Cognitive architectures such as SOAR [31], ACT-R [32], OpenCog [33] etc. provide frameworks for modelling the full spectrum of cognitive functions: perception, memory, learning, and reasoning. These systems often combine symbolic modules with probabilistic learning systems, making them well-suited for agentic behaviour.

By integrating these conflict-resolution strategies, hybrid systems can leverage the interpretability of symbolic reasoning while retaining the adaptability and pattern-recognition strength of sub symbolic models.

Decision-making under uncertainty

Reasoning is not sufficient without effective decision-making. Agentic AI systems must make autonomous choices based on goals, expected outcomes, and risks.

Utility-based decision-making

Many agent architectures use **expected utility maximization** as a guiding principle, where agents:

- Assign utilities (rewards) to outcomes.
- Evaluate actions based on expected return.

This is formalized in decision-theoretic models and used extensively in reinforcement learning [4].

Bounded rationality and heuristics

In real-world environments, computational constraints prevent exhaustive reasoning. Agentic systems may thus rely on:

- Heuristics: Fast, approximate strategies [34].
- Anytime algorithms: That improve performance with more time.
- Satisficing: Choosing "good enough" rather than optimal solutions.

This aligns with the bounded rationality model [35], where agents optimize within the constraints of limited information and computation.

Key takeaways

This chapter explores how reasoning and decision-making underlie the autonomy and intentionality of agentic AI systems. While symbolic reasoning offers clarity and structured knowledge manipulation, statistical reasoning offers adaptability and robustness. Their integration in hybrid systems paves the way toward more powerful and trustworthy agents.

Feature	Symbolic Reasoning	Statistical Reasoning	Hybrid Systems
Interpretability	High	Low	Medium
Data Requirement	Low	High	Medium
Flexibility	Low	High	High
Generalization	Weak	Strong (with enough data)	Strong
Commonsense Reasoning	Strong	Weak	Medium to Strong

Table 7.1: Comparison across different reasoning systems.

Understanding and modelling these processes is essential for building AI agents that can act responsibly, learn continuously, and adapt to complex human environments.

Chapter 8: Uncertainty, Confidence, and Reasoning

For agents to act robustly in dynamic and unpredictable environments, they must reason under uncertainty. This involves quantifying their confidence in their beliefs and decisions, reflecting on their knowledge gaps, and adapting behaviour accordingly. Uncertainty estimation and meta-cognition (where agents reason about their own reasoning) are critical for developing safe, reliable, and adaptive agentic AI systems. In this chapter, we explore probabilistic models for representing uncertainty, discuss active learning strategies that enable agents to gather informative data, and examine self-reflective processes that let agents monitor and improve their own reasoning.

Probabilistic models in agentic decisions

Probabilistic reasoning provides a mathematical framework for modelling uncertainty in perception, knowledge, and decision-making. Unlike deterministic systems, probabilistic agents maintain beliefs about the world as distributions over possible states, updating these beliefs as new evidence arrives.

Bayesian decision theory

Bayesian frameworks enable agents to calculate posterior distributions over hypotheses using Bayes' theorem, integrating prior beliefs with observed data:

$$p(\theta|D) = \frac{p(D|\theta)p(\theta)}{p(D)}$$

This allows agents to:

- **Estimate model uncertainty**: How confident is the agent in its model parameters?
- **Estimate predictive uncertainty**: How confident is the agent about future outcomes?

Bayesian reasoning supports expected utility maximization: choosing actions that maximize expected reward given current beliefs. Agents can trade off exploration (seeking information) and exploitation (pursuing immediate goals) explicitly.

Probabilistic graphical models

Graphical models such as Bayesian Networks or Conditional Random Fields allow agents to represent dependencies among variables, enabling efficient inference and structured reasoning. They support tasks like:

- **Causal reasoning**: Inferring the effects of interventions.
- **Contextual reasoning**: Updating beliefs given partial observations.

Active learning and exploration

An agent that only passively receives data is unlikely to learn efficiently, especially in complex environments. Active learning equips agents with strategies to actively select data points that maximize information gain, reducing sample complexity.

Principles of active learning

Agents can quantify **epistemic uncertainty** (uncertainty due to lack of knowledge) and use it to prioritize which data points to query or which actions to take. Strategies include:

- **Uncertainty sampling**: Selecting points with highest model uncertainty.
- **Expected model change**: Choosing queries that will most change the model parameters.
- **Information gain maximization**: Selecting actions that maximize expected reduction in entropy.

Exploration

In reinforcement learning, agents face the exploration-exploitation dilemma. Techniques for exploration include:

- **∊-greedy policies**: Randomly selecting non-optimal actions with probability ∊.
- **Upper Confidence Bounds (UCB)**: Choosing actions with the highest upper confidence bound on expected reward.
- **Thompson sampling**: Sampling model parameters from the posterior and choosing actions based on the sampled model.

These methods allow agents to systematically reduce uncertainty about the environment, leading to faster and more reliable learning.

Self-reflection

Self-reflection refers to an agent's capacity to monitor and evaluate its own internal processes, knowledge, and decision strategies. This is central to agentic AI, as it underpins adaptability, self-improvement, and trustworthiness.

Monitoring performance

An agent can maintain meta-level estimates of its performance metrics (e.g., prediction accuracy, uncertainty calibration) to detect when:

- The environment has changed (concept drift).
- Its predictions are consistently unreliable.
- A knowledge gap exists requiring new data or updated models.

Reflective adjustment

Once performance monitoring identifies issues, the agent can adjust:

- Update or retrain its model.
- Seek new training data (active learning).
- Modify planning or reasoning strategies.

This reflective loop mirrors human metacognition: our ability to recognize and correct errors in our thinking.

Meta-cognition

Meta-cognition extends self-reflection to include reasoning about one's own reasoning. It allows agents to explicitly represent and manipulate knowledge about their own cognitive states, plans, and capabilities.

Following are the components of meta-cognition:

- **Meta-knowledge**: Knowledge about the agent's own knowledge (e.g., *I know that I don't know the answer to this question.*)
- **Meta-reasoning**: Processes for controlling and allocating cognitive resources (e.g., deciding whether to spend more time deliberating).
- **Meta-learning**: The ability to improve one's learning strategies over time.

Cognitive architectures like Soar and ACT-R implement meta-cognitive components that let agents evaluate and modify their own problem-solving strategies [31], [32]. Recent advances in meta-learning algorithms also allow agents to adapt their learning processes dynamically across tasks [20].

Key takeaways

An agentic system capable of robust autonomous behaviour must unify:

- Uncertainty estimation to measure confidence.
- Active exploration to reduce epistemic uncertainty.
- Self-reflection to monitor performance and detect failure modes.
- Meta-cognition to adapt strategies and improve learning over time.

These components together enable agents to act responsibly in dynamic environments, adapt to unforeseen situations, and earn the trust of human collaborators.

Below is a schematic flow of the chapter's core components into a potential agentic AI application:

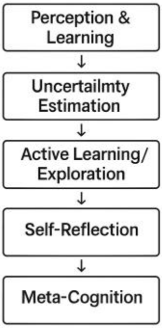

Figure 8.1: Schematic of integrating uncertainty, reflection, and meta-cognition.

Understanding and modelling uncertainty, confidence, and reflective reasoning are essential to build agentic AI systems that act robustly, learn effectively, and adapt responsibly. By combining probabilistic models with mechanisms for self-reflection and meta-cognition, agents can navigate real-world complexity with greater competence and trustworthiness.

PART III: Architectures and Systems

As artificial intelligence systems evolve from task-specific tools to autonomous agents, their architectural design becomes increasingly critical. Building agentic AI requires integrating diverse subsystems—perception, memory, reasoning, planning, learning, and action—into cohesive, adaptive entities capable of goal-directed behaviour in complex environments. This integration poses technical, conceptual, and philosophical challenges far beyond those addressed by conventional machine learning pipelines.

In earlier chapters, we explored the philosophical foundations and technical building blocks of agentic systems. We now turn to the architectures that operationalize these components into functioning, embodied agents. These architectures serve as blueprints for cognition, encoding how agents perceive their environment, represent internal knowledge, make decisions, reflect on past experiences, and coordinate behaviours to achieve evolving goals.

This part of the book examines both symbolic and sub-symbolic systems, as well as hybrid cognitive architectures that attempt to bridge the gap between explicit reasoning and learned representation. We also explore modular systems designed for robustness and scalability, drawing inspiration from neuroscience, robotics, and human cognition.

A central theme in this section is the idea of coherence across time and context. Agentic AI systems are not mere pipelines; they must maintain long-term goals, integrate new information with prior beliefs, and adapt their behaviour while preserving internal consistency. Achieving this requires recurrent, memory-augmented, and self-modifying architectures (systems that not only act in the world but evolve based on their own experiences and internal evaluations).

Throughout this part, we will also emphasize real-world deployment. Architectures must contend with noisy sensors, ambiguous goals, conflicting feedback, and limited data (conditions common in robotics, autonomous systems, decision-support tools, and digital assistants). As such, this section connects theory to practice, illustrating how abstract concepts manifest in practical, implemented systems.

Part III covers chapters to build a comprehensive picture of what it takes to design AI systems that are not only intelligent, but agentic, capable of autonomous, adaptive, and accountable behaviour across domains.

Let us now begin with the next chapter, where we lay the groundwork for understanding the strengths and limitations of two dominant paradigms in AI systems design.

Chapter 9: Architectural Paradigms for Agentic AI

As AI systems increasingly embody autonomy, persistence, and decision-making, architectural choices become foundational. An architecture defines not just how an AI system is implemented, but how it thinks, plans, learns, and adapts. In agentic AI, where the goal is to build entities that act with intentionality and awareness of their goals, beliefs, and contexts, architecture determines the boundaries of capability.

This chapter surveys the architectural paradigms central to agentic AI, tracing the evolution from symbolic logic systems to hybrid and language model-driven agents. Specifically, we focus on:

- The **Belief-Desire-Intention (BDI)** model, a long-standing formalism for rational agency.
- **Cognitive architectures** like SOAR and ACT-R that emulate human cognition.
- Emerging **LLM-based agent frameworks** that leverage foundation models as flexible cognitive substrates.
- The fundamental dichotomy and synergy between **symbolic and sub-symbolic architectures**.

These paradigms are not mutually exclusive. Increasingly, modern agentic systems blend their strengths into hybrid cognitive architectures, combining interpretability and logical rigor with adaptive learning and generalization.

Belief-Desire-Intention (BDI)

The BDI architecture is one of the earliest formal models of agency, derived from philosophical theories of practical reasoning [36]. In this framework, an agent is defined in terms of:

- **Beliefs** (B): Information the agent has about the world (may be uncertain or incomplete).
- **Desires** (D): Goals or states of the world that the agent wants to bring about.
- **Intentions** (I): Commitments to specific plans of action that the agent actively pursues.

A BDI agent continuously cycles through perception, reasoning, and action:

- **Perception**: Update beliefs based on new observations.
- **Deliberation**: Evaluate which desires to pursue based on current beliefs.
- **Intention Formation**: Commit to specific goals and plans.
- **Execution**: Act in the environment while monitoring outcomes and revising intentions if needed.

This approach mirrors human-like rationality, with mechanisms for goal prioritization, plan revision, and intentional action selection.

Prominent BDI frameworks include:

- PRS (Procedural Reasoning System): Early BDI system used in autonomous agents [37].

- Jason: A Java-based implementation widely used in multi-agent simulations [38].
- BDI in Robotics: Used for mission planning in autonomous drones and Mars rovers.

While BDI offers clarity and interpretability, its symbolic nature limits adaptability in uncertain or novel environments without extensive rule engineering.

Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is a powerful technique that combines two key steps:

- Retrieval of relevant information,
- Generation of natural language responses.

Instead of relying solely on an AI model's internal knowledge (which may be outdated or incomplete), RAG dynamically pulls up-to-date context from external sources, such as document databases, APIs, or knowledge bases, before producing its answer.

In agentic AI, RAG is crucial for enabling agents to make factually grounded, context-aware decisions. By retrieving relevant data in real time, agents can work with current information, handle niche queries, and avoid hallucinations.

Examples:

- **Customer Support:** An AI agent retrieves troubleshooting guides and policy documents before generating a personalized reply to a customer inquiry.

- **Healthcare:** A clinical assistant agent queries a medical knowledge base to provide a doctor with evidence-based treatment suggestions.
- **Business Intelligence:** A financial research agent fetches the latest earnings reports and market data before writing an investment summary.
- **Legal Research:** An AI paralegal retrieves case law from a database and synthesizes it into an argument draft for a lawyer.

Cognitive architectures

Cognitive architectures aim to model the full range of human cognitive processes, from perception and memory to problem-solving and learning. They provide blueprints for building general intelligent agents.

Cognitive architectures offer deep explanatory power and psychological realism, making them ideal for agents interacting with humans. However, they often struggle with open-world adaptability and scalability.

Architecture	Strengths	Limitations
SOAR	Structured problem-solving, hierarchical goals	Symbolic, needs task-specific rules
ACT-R	Human-like memory, validated against experiments	Limited scalability
CLARION	Models both implicit and explicit processes	Less mature tooling

Table 9.1: Comparison of cognitive architectures.

SOAR

- Developed at Carnegie Mellon University, SOAR (State, Operator, And Result) is a production system architecture.
- Agents maintain a working memory, apply production rules to generate candidate actions, and use chunking to learn new rules.
- SOAR supports goal decomposition, where complex tasks are broken into subtasks hierarchically.

ACT-R (Adaptive Control of Thought - Rational)

- ACT-R models cognition as a set of modules (e.g., declarative memory, procedural memory, perception, motor control) interacting through a central buffer.
- Learning occurs via declarative chunking and reinforcement learning.
- ACT-R has been extensively validated against psychological experiments, making it suitable for modelling human-agent interaction, education, and decision-making.

CLARION

- Focuses on dual-process cognition: implicit (bottom-up) vs. explicit (top-down) processes.
- Useful in modelling motivation, emotion, and social behaviour.

LLM-based agent frameworks

With the advent of foundation models like GPT-4, Claude, and Gemini, a new architectural paradigm has emerged: LLM-centric agentic systems. These agents use pre-trained language models as the central reasoning engine, surrounded by specialized modules for memory, tools, and planning. Examples:

- AutoGPT / BabyAGI: Use LLMs to recursively decompose goals and generate task sequences.
- LangChain Agents: Wrap LLMs with tools, memory stores, and search capabilities.
- Agentic RAG (Retrieval Augmented Generation): Allows AI agents to fetch real-world data to enhance knowledge and accuracy with the help of internal/external knowledge base.
- ReAct (Reasoning and Acting): Integrates chain-of-thought reasoning with real-time tool use [5].

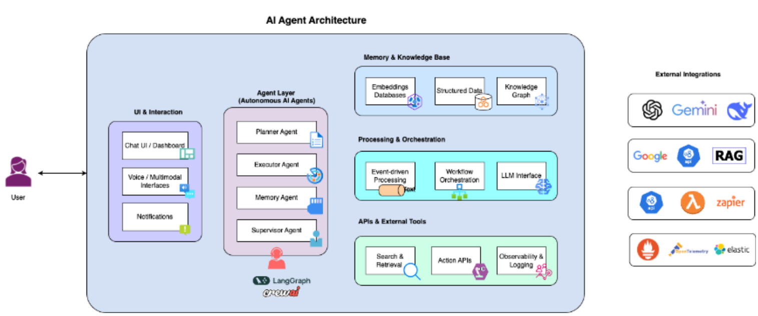

Figure 9.1: LLM-based agentic AI architecture framework example (source: GoPenAI).

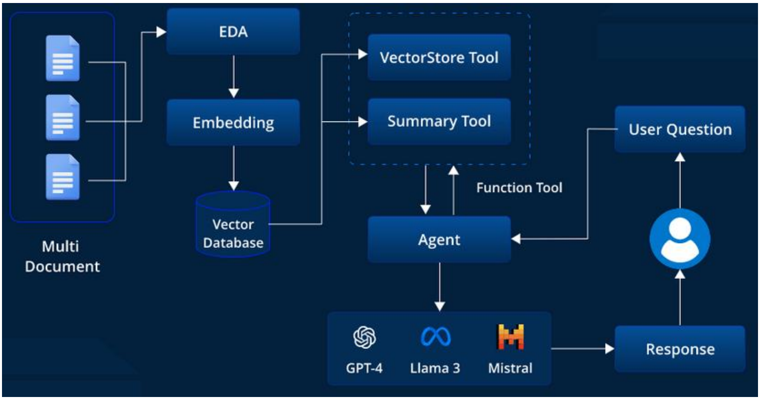

Figure 9.2: Agentic RAG example (source: LeewayHertz).

The following code snippet shows a simple LangChain-based agent for answering questions.

```
python

from langchain import OpenAI, LLMChain,
PromptTemplate

template = """You are an autonomous assistant.
User Question: {question}
Answer:"""
prompt = PromptTemplate(input_variables=["question"],
template=template)
llm = OpenAI(model="gpt-4")
chain = LLMChain(llm=llm, prompt=prompt)

response = chain.run("What is the capital of
France?")
print(response)
```

Topologies

Agentic AI systems require a flexible, resilient architecture that reflects the ways humans, both individually and in groups, approach complex problem-solving. These agents may operate independently or collaborate within networks of varying configurations.

- **Single-Agent Architecture**: Consists of a single autonomous agent handling all tasks.
- **Multi-Agent Systems (MAS)**: Contain two or more agents that may be organized into different topologies:
    - **Vertical (Supervised) Architecture**: A central "supervisor" agent coordinates interactions when agents have conflicting objectives. The supervisor prioritizes system-wide goals and negotiates compromises. For example, if buyer and seller agents cannot agree on a price, the supervisor mediates the deal.
    - **Network Architecture**: Agents communicate directly with each other to achieve shared MAS goals. For instance, autonomous vehicles may exchange information to safely navigate and park in tight spaces, balancing both individual and collective safety objectives.

Characteristics

An agentic AI system can be defined by the following core capabilities:

- **Autonomy**: Initiates and completes tasks with little to no human oversight, allowing for greater operational flexibility and efficiency.
- **Reasoning**: Analyses context to make well-informed decisions and exercise sound judgment.
- **Reinforcement Learning**: Adapts and improves over time by interacting with its environment and learning from feedback.

- **Language Comprehension**: Understands and follows complex instructions to ensure clear communication and precise execution, including zero-shot or few-shot learning using natural language.
- **Workflow Optimization**: Manages and streamlines multi-step processes for maximum efficiency.
- **Flexible planning** via language-based reasoning.
- **Tool Calling**: Interfaces with APIs, databases, and environments.

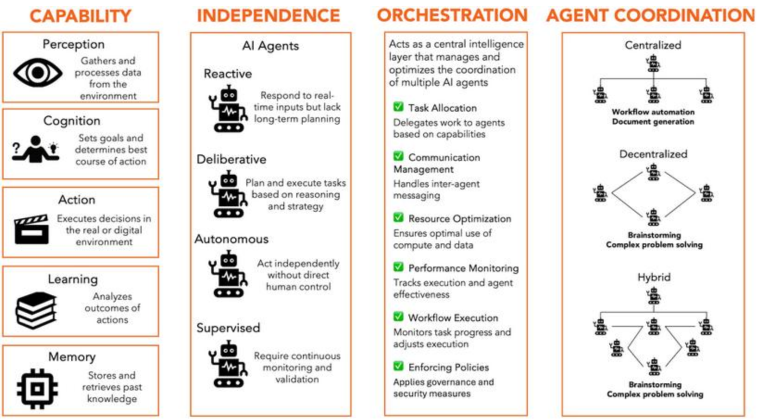

Figure 9.3: Agentic AI architecture framework example (source: LinkedIn).

Limitations

However, these systems also face challenges:

- **Hallucination**: LLMs may generate plausible but incorrect outputs.
- **Lack of grounding**: Text-based understanding may lack embodied context.

- **Poor long-term memory**: LLMs struggle with persistent internal states.
- **Self-contradictions**: LLM-based agents struggle in maintaining long-term goals and self-consistency.

Framework	Agent Type	Tools	Sequential Workflows	Dynamic Workflows	Memory	Guardrails
Langchain	✅ Regular agent	✅ Out of the box	✅	❌	✅	❌
LangGraph	✅ Regular agent with state	✅ Same as Langchain	✅	✅ Custom graphs can be built	✅ State	❌
Crew AI	✅ Planning and Manager agent	✅ Langchain and LlamaIndex	✅	✅ Hierarchical	✅ Long-term, short-term, entity, context	❌
Swarm	✅ Regular agent with state	❌ No out-of-the-box	✅	❌ Build using handoffs	✅ State and context variables	❌
LlamaIndex	TBD	TBD	TBD	TBD	TBD	TBD
MSFT Autogen	TBD	TBD	TBD	TBD	TBD	TBD

Figure 9.4: Comparison of LLM-based agentic frameworks (source: Swapan Rajdev, Haptik).

While LLM-based agent frameworks such as LangChain, Auto-GPT, CrewAI, and BabyAGI have demonstrated impressive capabilities in autonomous task execution, they still face notable challenges in maintaining long-term goals and ensuring self-consistent behaviour. Due to their reliance on large transformer-based models with fixed context windows, these agents often lose track of prior reasoning steps or previously established objectives, especially in extended interactions or multi-session scenarios.

Furthermore, without robust memory architectures or explicit goal representation modules, they can produce decisions that are locally coherent but globally inconsistent.

This lack of persistent self-consistency hinders their suitability for tasks requiring extended temporal planning, complex dependencies, or alignment with evolving objectives over weeks or months. These issues connect directly to the broader challenges of goal stability, catastrophic forgetting, and reflective agency discussed in Part V, underscoring the need for integrated memory, meta-reasoning, and adaptive planning mechanisms in next-generation LLM-based agents.

Despite these, LLM agents have catalysed a new wave of agentic applications, from software engineering to autonomous research assistants.

Symbolic and sub-symbolic architectures

At the heart of AI lies the divide, and potential synthesis, between symbolic and sub-symbolic paradigms.

Approach	Pros	Cons
Symbolic	Transparent, interpretable	Rigid, poor generalization
Sub-symbolic	Adaptive, scalable	Opaque, ungrounded reasoning
Hybrid	Best of both worlds	Complex to implement and debug

Table 9.2: Pros and cons of symbolic and sub-symbolic architectures.

Symbolic AI

- Based on explicit rules, logic, and knowledge representation.
- Allows transparent reasoning, modularity, and verifiability.

- Examples: BDI agents, knowledge graphs, planning systems.

Sub-symbolic AI

- Based on neural networks, vector representations, and gradient descent.
- Enables pattern recognition, generalization, and learning from raw data.
- Examples: Deep learning models, embeddings, RNNs.

Hybrid architectures

Increasingly, agentic systems combine both. Such systems:

- Use symbolic planners for high-level goals and neural networks for perception.
- Combine logical inference with embedding-based retrieval.
- Integrate language models with knowledge bases.

One of the challenges in hybrid cognitive architectures is managing situations where symbolic reasoning (explicit, rule-based logic) and sub-symbolic reasoning (statistical or neural-based pattern recognition) produce conflicting recommendations.

For example, in an autonomous medical diagnostic assistant, the symbolic module might follow clinical guidelines that recommend a particular test based on symptom combinations. Meanwhile, the sub-symbolic neural model trained on historical patient data might predict that the same test has a low probability of yielding useful results for this specific patient, based on subtle data patterns invisible to the symbolic rules.

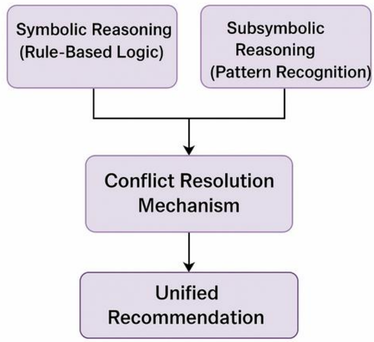

Figure 9.5: Conflict-resolution flow between symbolic and sub-symbolic modules.

Hybrid architectures resolve such conflicts through mechanisms like:

- **Meta-reasoning layers**, which weigh the confidence of each subsystem's output.
- **Arbitration rules**, which prioritize certain decision sources under specific contexts (e.g., symbolic rules override when legal compliance is at stake).
- **Consensus-building algorithms**, which synthesize outputs into a unified recommendation.

Key takeaways

Agentic AI demands more than just learning or prediction. It requires structured cognition, adaptive decision-making, and persistent agency. Architectural choices shape what kinds of behaviour an AI agent can produce, and how it adapts over time.

This chapter explored:

- The BDI model as a foundation for intentional behaviour.
- Cognitive architectures as blueprints for human-like agency.
- The rise of LLM-based agents as flexible, language-driven decision-makers.
- The ongoing synthesis of symbolic and sub-symbolic approaches in hybrid systems.

As we move forward in building embodied, task-aware, and socially aligned AI, architectural foundations will remain central. In the next chapter, we examine how agents communicate with their environment and users through multimodal interaction, memory updates, and planning loops.

Chapter 10: Open-ended Learning Agents

In the trajectory toward general intelligence, open-ended learning represents a critical milestone. Unlike conventional AI systems that operate within fixed environments and task boundaries, open-ended learning agents continuously evolve, acquiring new knowledge, expanding their skill sets, and refining their behaviour in unstructured, dynamic, and unpredictable contexts.

These agents are not limited by static reward functions, predefined datasets, or narrow problem domains. Instead, they are designed to self-improve, explore, generalize, and even develop new goals or representations of the world. This chapter explores the mechanisms behind open-ended learning, focusing on self-improvement and evolutionary computation, two foundational paradigms that empower agents to transcend traditional boundaries.

Open-ended learning refers to the capability of an agent to autonomously and continually acquire novel and increasingly complex behaviours or knowledge without an external endpoint or target task. The process is **non-stationary**, **unbounded**, and often **self-driven**.

Feature	Description
No fixed objectives	Unlike supervised learning, there's no single objective function.
Lifelong learning	Knowledge is accumulated across a sequence of experiences or tasks.
Self-generation of goals	Agents can propose and pursue their own learning objectives.
Continual expansion of capacity	New skills are built upon prior capabilities.

Table 10.1: Key characteristics of open-ended learning.

Examples:

- **XLand** (DeepMind): Agents learn diverse skills in procedurally generated 3D environments, guided by meta-rewards and self-discovery.
- **GPT** (OpenAI): Though not fully autonomous learners, these models demonstrate scaling-based emergent behaviour through unsupervised learning.
- **CICERO** (Meta): Blends dialogue planning, strategic reasoning, and memory, suggesting possible future integration with open-ended reasoning agents.

This paradigm mimics natural evolution and development, where organisms adapt, evolve, and develop increasing complexity in an ever-changing environment.

Self-improvement mechanisms

Self-improvement is the capacity of an AI agent to monitor, evaluate, and enhance its own performance. While it shares overlap with continual learning, self-improvement emphasizes autonomy: the agent actively identifies deficiencies and takes steps to overcome them.

To enable self-improvement, an agent typically requires:

- Self-monitoring components (e.g., metacognition or introspective modules)
- Performance estimators (e.g., confidence scoring or prediction error analysis)
- Adaptation modules (e.g., fine-tuning, transfer learning, or architecture search)
- Memory systems to retain experiences and evaluate progress over time

Figure 10.1: High-level Self-Improvement Loop

Different techniques are used to enable self-improvement:

- **Self-Supervised Learning**: Enables learning from raw data without explicit labels, often through contrastive or predictive objectives (e.g., SimCLR, BYOL).
- **Curriculum Learning**: Agents progressively tackle more complex tasks, with the difficulty increasing based on the agent's current skill level.
- **Meta-Learning**: The agent improves its learning process itself, often referred to as "learning to learn" [20].
- **Uncertainty-based Exploration**: Agents prioritize experiences or tasks where their predictive uncertainty is highest (e.g., Bayesian or ensemble-based exploration).
- **Intrinsic Motivation**: Inspired by human curiosity, intrinsic motivation mechanisms (e.g., novelty search, empowerment, predictive error minimization) push agents to explore unfamiliar situations.

Evolutionary agents

Inspired by biological evolution, evolutionary computation involves generating populations of agent variants, evaluating them according to a fitness function, and selecting the best performers to "reproduce" via mutation and crossover.

Evolutionary algorithms work in the following way:

- **Initialization**: Generate an initial population of agents (e.g., neural networks).
- **Evaluation**: Each agent interacts with an environment and receives a fitness score.
- **Selection**: High-performing agents are selected to breed.
- **Mutation/Crossover**: New agents are generated through stochastic mutation or recombination.
- **Repeat**: The cycle continues across generations.

The following code snippet provides a minimal evolutionary loop for improving an agent's score.

```
python

import random

def fitness(agent):
    return sum(agent)  # Dummy fitness: sum of
parameters

population = [[random.random() for _ in range(3)] for
_ in range(5)]

for generation in range(10):
    population.sort(key=fitness, reverse=True)
    # Keep top 2 and mutate
    population = population[:2] + [
```

```
        [x + random.uniform(-0.1, 0.1) for x in
population[0]],
        [x + random.uniform(-0.1, 0.1) for x in
population[1]]
    ]
    print(f"Gen {generation}, Best fitness:
{fitness(population[0])}")
```

Evolutionary agents are particularly powerful for:

- Architecture search (e.g., evolving neural network topologies),
- Robotics and control (where gradient-based optimization is challenging),
- Emergent behaviour modelling.

Framework	Description
NEAT (Neuro-Evolution of Augmenting Topologies)	Evolving both weights and network structures
ES (Evolution Strategies)	Black-box optimization using population sampling
POET (Paired Open-Ended Trailblazer)	Co-evolves environments and agents for lifelong learning (Wang et al., 2019)
MAP-Elites	Maintains a diverse set of high-performing solutions across a feature space

Table 10.2: Notable frameworks and approaches of evolutionary agents.

Evolutionary systems, especially when paired with open-ended environment generators, can lead to emergent intelligence, where agents develop unexpected capabilities, strategies, or forms of collaboration.

Combining self-improvement with evolution

Modern approaches blend evolutionary algorithms with self-improving mechanisms to create more robust and generalizable learning systems.

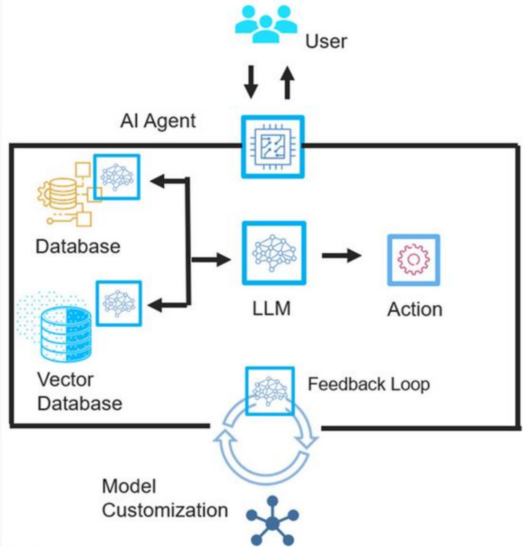

Figure 10.2: Self-Improving agentic AI (source: Cadence).

For example, OpenAI's Evolution Strategy + Gradient Descent approach use evolutionary strategies to search large-scale hyperparameter spaces and use gradient-based methods for local fine-tuning.

Similarly, in Population-Based Training (PBT), agents train in parallel, periodically exchanging weights and hyperparameters. This approach combines exploration (mutation) and exploitation (gradient updates).

Model Context Protocol (MCP)

Model Context Protocol (MCP) is an emerging standard that enables AI systems, particularly agentic AI, to interact with external tools, data sources, and APIs in a structured, secure way. Instead of hardcoding every possible integration, MCP provides a universal protocol that allows AI agents to dynamically discover and use available resources at runtime.

In the context of agentic AI, MCP is critical because it gives autonomous agents the ability to act intelligently across multiple systems. For example, an agent can request data from a company's database, trigger workflows in a project management tool, or access real-time market information, all without manual reconfiguration. This protocol abstracts away technical complexity, allowing AI to focus on reasoning and decision-making while relying on MCP to handle safe, standardized communication with the outside world.

Example: Imagine an AI research assistant tasked with compiling a weekly market analysis report.

1. **Discovery:** The agent uses MCP to discover available tools, such as a stock market data API, a company news database, and a spreadsheet editor.
2. **Planning:** Based on its goal, it decides to fetch the latest market data, summarize key news, and generate charts.
3. **Execution:** Through MCP, the agent securely queries the market API, retrieves relevant data, writes summaries into a spreadsheet, and emails the final report to the team.

This entire process happens without manual API wiring. MCP standardizes how the agent communicates with each system, making the workflow more scalable and adaptable.

Key takeaways

Open-ended learning marks a departure from task-constrained AI to systems capable of autonomous evolution, goal generation, and continual self-improvement. From self-supervised agents that fine-tune their skills to evolutionary systems that co-develop their architectures and behaviours, the path toward general agentic intelligence increasingly relies on open-endedness as a core principle. However, there are certain challenges as well.

Challenge	Description
Catastrophic forgetting	As agents learn new tasks, they forget old ones.
Evaluation difficulties	No fixed goal makes progress hard to quantify.
Computational demands	Evolutionary and lifelong agents require massive compute.
Stability	Balancing exploration with convergence is non-trivial.

Table 10.3: Challenges in Building Open-Ended Learning Agents.

Despite these challenges, progress in scaling laws, meta-learning, and unsupervised world modelling continues to enable more sophisticated open-ended learning agents.

In the next chapter, we explore how agents interact with other agents and humans through cooperation, communication, and competition in multi-agent systems.

Chapter 11: Multi-Agent Systems and Social Intelligence

Human intelligence has evolved in inherently social contexts. We collaborate in teams, negotiate resources, build shared knowledge, and engage in strategic competition. If agentic AI is to achieve general-purpose intelligence or coexist meaningfully with humans, it must exhibit social intelligence, the ability to perceive, interpret, and influence other agents.

In this chapter, we examine multi-agent systems (MAS), environments where multiple AI agents (and possibly humans) interact. These systems require AI agents not only to pursue their own goals, but also to understand the intentions, beliefs, and actions of others. We explore the mechanics of **cooperation**, **competition**, and **communication** in MAS and how game theory and agent **negotiation** frameworks empower agents with social reasoning capabilities.

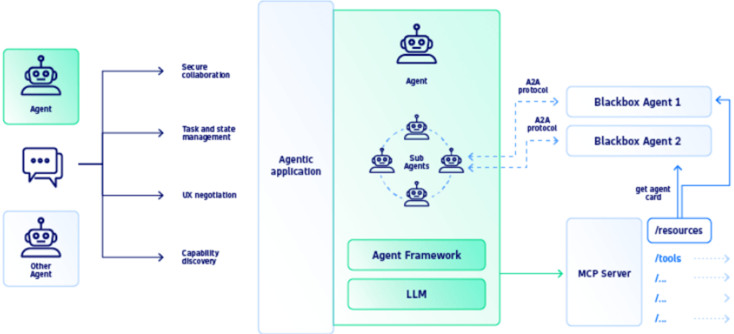

Figure 11.1: Example of agent-to-agent architecture (source: Dynatrace).

Multi-agent system (MAS)

A multi-agent system (MAS) is a computational system composed of multiple interacting agents, which may be:

- Collaborative (shared or aligned goals),
- Competitive (conflicting goals),
- Mixed motive (both cooperation and competition).

Domain	Example Use
Robotics	Drone swarms, warehouse robot coordination
Economics	Market simulations, trading bots
Transportation	Multi-agent traffic routing, ride-sharing optimization
Gaming	Multiplayer AI, strategy games
Crisis response	Multi-agent planning for disaster scenarios
Military simulations	Adversarial agent planning

Table 11.1: Real-world applications of MAS.

These agents can be heterogeneous (different designs, capabilities) or homogeneous, and interactions may be:

- Simultaneous or sequential,
- Symmetric or asymmetric,
- Centralized or decentralized.

We are entering a three-stage evolution of Enterprise AI agents.

- **Stage 1**: *Single Agents (Specialized Contributors)*: In the initial phase, highly specialized agents focus on well-defined tasks within specific industries. They deliver personalized insights, precise generative outputs, and workflow enhancements, improving customer interactions, marketing efforts, and operational efficiency. By handling routine processes with speed and accuracy, these agents pave the way for broader enterprise AI adoption.

- **Stage 2**: *Collaborative Agents (Coordinated Teams)*: The second stage introduces an orchestrator agent that coordinates multiple specialized agents within a single organization, similar to a restaurant manager overseeing chefs, servers, and cleaners to ensure a seamless dining experience. The orchestrator aligns tasks and priorities toward shared business objectives.
- **Stage 3**: *Agent Ecosystems (Ensemble Orchestrators)*: In the final stage, advanced agent-to-agent (A2A) collaboration spans across organizations, creating new interaction models such as business-to-agent (B2A) and business-to-agent-to-customer (B2A2C). Here, AI agents act as intermediaries, enabling cross-industry cooperation, complex negotiations, risk management, and conflict resolution, driving personalized, high-value collaboration at scale.

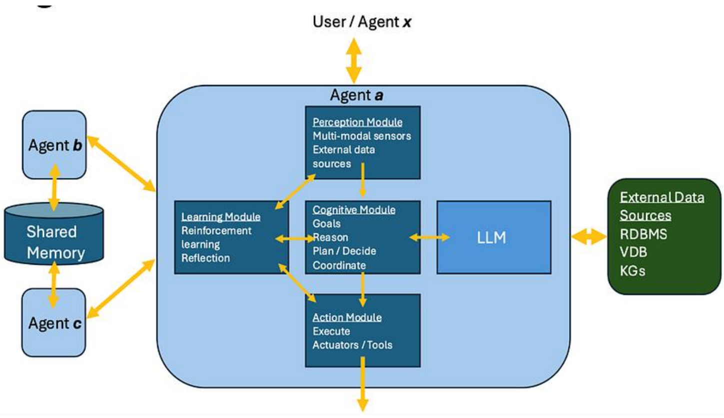

Figure 11.2: Example of MAS configuration (source: UC Berkeley).

Property	Description
Decentralization	No single agent has global control.
Partial Observability	Agents only have local views.
Communication	Agents may share beliefs, goals, or observations.
Coordination	Necessary to avoid conflict and improve joint performance.
Adaptivity	Agents must adapt to others' changing strategies.

Table 11.2: Characteristics of Multi-Agent Systems.

Cooperation

Cooperation in MAS can happen in two ways:

- **Centralized**: A controller coordinates all agents (e.g., multi-robot planning).
- **Decentralized**: Agents must coordinate autonomously.

Agents may coordinate by sharing:

- Observations
- Goals
- Beliefs
- Strategies

Multi-Agent Reinforcement Learning (MARL) often includes message passing modules or shared attention to facilitate this.

Mechanism	Description
Joint Intentions	Agents form shared goals and track mutual commitments.
Task Allocation	Roles or subtasks are distributed optimally.

Shared Policies	Agents follow a common strategy, e.g., distributed reinforcement learning.
Multi-Agent Planning	Agents build plans that consider interdependencies.

Table 11.3: Mechanisms for cooperation.

Competition

Examples of Competitive MAS include:

- Multi-player games (e.g., StarCraft, Diplomacy, Poker),
- Auctions (e.g., bidding strategies),
- Cybersecurity simulations (attacker-defender models),
- Financial markets (trader bots competing on prices).

Agents often compete over:

- Resources,
- Influence,
- Survival,
- Market share.

In competitive scenarios, agents:

- Anticipate opponents' moves (e.g., via opponent modelling).
- Use deception or bluffing.
- Learn robust policies that generalize against unseen adversaries.

The following code snippet shows a very basic multi-agent resource-sharing simulation.

```
python

agents = {"A": 10, "B": 8}
```

```
resource_pool = 5

# Cooperative strategy
for agent in agents:
    agents[agent] += resource_pool / len(agents)
print("After cooperation:", agents)

# Competitive strategy (winner takes all)
agents = {"A": 10, "B": 8}
winner = max(agents, key=agents.get)
agents[winner] += resource_pool
print("After competition:", agents)
```

Negotiation

In shared environments, agents often negotiate over:

- Task roles or responsibilities
- Resource allocations
- Agreements or contracts

Some of the common negotiation strategies used are:

- Tit-for-Tat: Cooperate initially, then reciprocate behaviour.
- Win-Stay, Lose-Shift: Maintain behaviour if successful.
- Utility Maximization: Model payoffs and maximize expected return.

Following negotiation protocols are popular in agentic systems:

- **Bilateral negotiation**: One-to-one interaction.
- **Auction mechanisms**: E.g., Vickrey auctions.
- **Contract nets**: Distributed task contracting.

Communication

Agents need mechanisms to signal, query, or coordinate. Communication may be of different kinds:

Mode	Example
Explicit	Natural language, symbolic messages
Implicit	Action-based cues, shared environment changes
Emergent	Agents develop protocols (e.g., via MARL)

Table 11.4: MAS communication and language.

Recent advancements in LLM-agent frameworks (e.g., AutoGPT, Voyager) enable natural language-based planning and negotiation, bringing us closer to human-agent collaboration.

Social intelligence in agents

Theory of Mind (ToM)

A key milestone in social intelligence is Theory of Mind, the ability to model what other agents believe, intend, or desire.

AI implementations of ToM are typically:

- **Zero-order**: Models only the environment.
- **First-order**: Models other agents' goals or actions.
- **Second-order and beyond**: Models what others believe about itself, etc.

This is foundational for:

- **Prediction**: Anticipating the next move of others.
- **Deception**: Misleading other agents strategically.
- **Trust and cooperation**: Estimating reliability or intentions.

Social learning

Agents in MAS can learn not only from the environment, but from each other. Techniques include:

- **Imitation Learning**: Mimicking observed behaviour.
- **Observational Learning**: Updating beliefs from others' outcomes.
- **Emergent Communication**: Developing shared languages or protocols.

Game theory and agentic decision-making

Game theory provides the mathematical foundation for analysing interactions between rational agents. The applications of game theory in AI include:

- Training AI agents in simulated economies.
- Optimizing supply chains with competing vendors.
- Coordinating traffic flows in multi-agent routing.

The core concepts involved are:

- **Nash Equilibrium**: No agent has incentive to deviate unilaterally.
- **Pareto Optimality**: No one can be better off without someone else being worse off.
- **Minimax Strategy**: Optimal play against a worst-case adversary.
- **Mechanism Design**: Designing rules to induce desirable outcomes.

Game Type	Example
Cooperative	Coalition formation, resource sharing

Competitive (Zero-sum)	Chess, Go
Mixed motive	Prisoner's Dilemma, Stag Hunt
Repeated games	Iterated games with memory or history
Bayesian games	Incomplete information about other player types

Table 11.5: Types of games.

The following code snippet simulates iterated Prisoner's Dilemma.

```
python

import random

def tit_for_tat(history):
    return history[-1][1] if history else "C"

def random_strategy(_):
    return random.choice(["C", "D"])

def play_game(strategy1, strategy2, rounds=5):
    payoff = {"CC": (3, 3), "CD": (0, 5), "DC": (5,
0), "DD": (1, 1)}
    history = []
    scores = [0, 0]
    for _ in range(rounds):
        move1 = strategy1(history)
        move2 = strategy2([(m2, m1) for m1, m2 in
history])
        scores[0] += payoff[move1 + move2][0]
        scores[1] += payoff[move1 + move2][1]
        history.append((move1, move2))
    return scores

print(play_game(tit_for_tat, random_strategy))
```

Key takeaways

Multi-agent systems represent a significant frontier in agentic AI. Social intelligence, manifested through cooperation, competition, negotiation, and communication, is essential for building agents that can function in complex, interactive environments.

As agentic AI matures, the ability to model, anticipate, and negotiate with other entities (human or machine) will be fundamental to building trustworthy, collaborative, and adaptable systems.

Chapter 12: Simulated and Embodied Agents

Agentic AI research spans a broad continuum from purely virtual agents that exist only in digital environments to physically embodied systems operating in the real world. Both forms, simulated agents and embodied agents, play a vital role in advancing the science and engineering of autonomy.

In this chapter, we will explore:

- **Robotics and embodied cognition**, where physical presence and sensory-motor interaction with the real world shape the agent's intelligence.
- **Virtual agents in games and simulations**, where large-scale, controlled, and repeatable environments allow rapid iteration and safe experimentation.

Robotics and embodied cognition

Embodied agents

An embodied agent is an autonomous system with a physical presence that perceives, reasons, and acts in the physical world. Examples include:

- Industrial robots (assembly arms, welding robots),
- Service robots (delivery bots, household assistants),
- Humanoid robots (social companions, bipedal walkers),
- Mobile robots (autonomous vehicles, exploration rovers).

Embodiment fundamentally changes the nature of AI. Unlike purely symbolic reasoning systems, embodied agents must deal with:

- Sensorimotor noise

- Physical constraints (gravity, friction, torque limits)
- Time delays and real-time control

Robotics as a platform for agentic AI

Robotics serves as a stress test for agentic AI systems:

- Real-world complexity: Agents must handle uncertainty, incomplete information, and hardware limitations.
- Goal-directed behaviour: Agents must plan over multiple time horizons while reacting to changing contexts.
- Adaptation: Agents must learn from failures and adapt to new physical conditions (e.g., a robot arm working with an unfamiliar object shape).

Embodied cognition theory

The embodied cognition hypothesis ([39]; [40]) argues that intelligence is deeply rooted in an agent's physical interactions with its environment. For AI, this means that:

- Cognitive processes are not just abstract computations but are grounded in sensorimotor experience.
- Learning is shaped by the structure of the body and the affordances of the environment.

An embodied agent's control loop follows a **perception–action cycle**. For example, a robot with a manipulator arm learns to grasp differently than one with suction tools, not because of different software, but because of physical embodiment constraints. Key components involved are:

- **Sensors**: Cameras, LiDAR, tactile sensors, microphones.
- **Perception**: Vision algorithms, SLAM (Simultaneous Localization and Mapping), object recognition.

- **Decision-making**: Planning algorithms, reinforcement learning policies.
- **Actuators**: Motors, servos, pneumatics for physical movement.

Following are the challenges in embodied AI:

Challenge	Example Scenario	AI Relevance
Real-time constraints	Drone avoiding obstacles mid-flight	Low-latency decision-making
Safety and reliability	Autonomous car in pedestrian zones	Risk-aware control
Transfer from simulation	Training in Gazebo, deploying on real robot	Domain adaptation
Energy efficiency	Warehouse robots running on limited battery	Planning with resource constraints

Table 12.1: Challenges in embodied AI.

Virtual agents in simulations

Simulations provide a safe, scalable, and controllable environment for developing and testing agentic AI. Virtual agents, such as NPCs (non-player characters) in games or simulated bots in virtual worlds, allow researchers to:

- Run millions of experiments without physical wear and tear.
- Test edge cases (rare events, adversarial conditions).
- Easily control environmental variables.

Simulation Environment	Primary Use Case	Agentic AI Role

OpenAI Gym / PettingZoo	RL benchmarks	Single & multi-agent learning
Unity ML-Agents	3D simulated environments	Visual navigation, interaction
CARLA	Autonomous driving simulation	Perception and planning
StarCraft II API	Real-time strategy AI	Multi-agent coordination
Habitat-Sim	Embodied navigation tasks	Vision-based planning

Table 12.2: Simulated platforms.

Emergent behaviours in simulated agents

Multi-agent simulations often reveal emergent behaviours: unexpected yet adaptive strategies. Examples:

- Cooperation in foraging tasks without explicit communication.
- Complex combat manoeuvres learned by game bots.
- Novel communication protocols emerging in MARL setups.

These emergent patterns often inspire real-world deployment strategies.

Virtual agents as physical system prototypes

Simulated agents serve as prototypes before deploying on real robots:

- Domain randomization helps agents generalize by exposing them to varied simulated conditions.
- Sim-to-real transfer bridges the gap between simulation and physical deployment.

- Hybrid training alternates between simulation and limited real-world trials to refine policies.

Key takeaways

Increasingly, research blurs the line between simulated and embodied agents:

- Digital twins replicate real-world systems in simulation for monitoring and optimization.
- Mixed-reality environments allow embodied robots to interact with simulated elements.
- LLM-driven virtual agents can later be embodied in robots without changing core cognition modules.

Simulated agents offer safety, speed, and scalability. Embodied agents bring the grounding and challenges of the physical world. Together, they form a complementary pair in the development of robust, adaptive, and socially capable agentic AI systems. Simulation accelerates innovation, embodiment tests reality. The future of agentic AI will likely feature tight integration between simulated training grounds and real-world deployments, ensuring agents are both efficient learners and effective actors.

AGENTIC AI

PART IV: Applications and Impact

Agentic AI, with its capacity for autonomous decision-making, goal-oriented behaviour, and adaptive learning, is not just a theoretical construct. It is increasingly shaping real-world systems. While the earlier parts of this book have focused on the foundational principles, philosophical underpinnings, and architectural paradigms of agentic intelligence, this section shifts the spotlight to practical applications and societal implications.

From precision healthcare to autonomous scientific discovery, agentic AI systems are beginning to influence domains where traditional automation was either infeasible or insufficient. These systems operate not only as passive tools but as collaborative partners, capable of forming strategies, negotiating with other agents, and adapting to fluid, high-stakes environments. The emerging potential is vast: self-directed robots in disaster relief operations, negotiation-capable AI in financial markets, or personal assistants that evolve their behaviour through sustained interactions with humans.

However, the deployment of agentic AI at scale raises complex questions about safety, transparency, ethics, and societal alignment. The very attributes that make these systems powerful—autonomy, adaptability, and persistence—also introduce novel risks, such as unpredictable behaviour, emergent strategies, or goal misalignment. As such, understanding the impact of agentic AI requires both domain-specific evaluations and cross-disciplinary policy frameworks.

In this part, we will explore how agentic AI systems are applied across diverse sectors, how their performance compares to traditional systems, and what challenges remain in integrating them responsibly into the human ecosystem. We will also examine the economic, ethical, and governance dimensions that will shape their trajectory in the coming decades.

The chapters ahead will delve into case studies, highlight innovation opportunities, and address the critical balancing act between harnessing agentic AI's potential and ensuring its safe, fair, and beneficial deployment.

Chapter 13: Agentic AI in the Real World

Agentic AI's transition from theoretical models and controlled research environments into real-world applications represents a defining moment in artificial intelligence. These systems are no longer confined to academic laboratories; they are actively influencing how robots navigate uncertain terrains, how students learn in adaptive classrooms, how manufacturing systems self-optimize, and how autonomous vehicles make split-second decisions in dynamic traffic. The defining characteristic in each of these applications is autonomy coupled with goal-directed adaptability, a capability that distinguishes agentic AI from static automation or narrow, rule-bound systems.

The emergence of advanced sensors, simulation environments, and large-scale computing infrastructure has accelerated the transition from conceptual frameworks to practical deployments. Autonomous delivery robots, adaptive tutoring systems, and self-driving vehicles are no longer isolated pilot projects. They are components of expanding ecosystems where agentic AI serves as the cognitive engine.

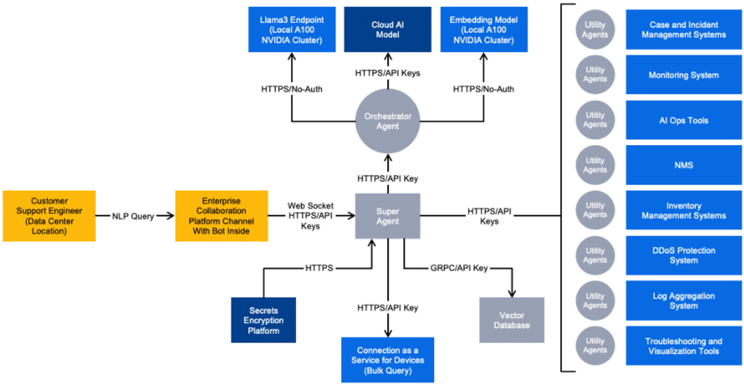

Figure 13.1: A real-world example of a multi-cloud, multi-model deployment of agentic AI inference (source: Equinix).

Yet, with these advances come pressing questions. How can we ensure that decision safety is maintained in unpredictable environments, such as urban traffic for autonomous vehicles or factory settings with human co-workers? How do we balance the autonomy of agentic AI with the need for human oversight in critical applications like medical robotics or adaptive educational platforms?

Domain	Example Applications	Benefits	Challenges
Robotics	Search-and-rescue drones, cobots, home assistants	Adaptability, multi-tasking, reduced supervision	Hardware cost, integration with human teams
Education	ITS, conversational tutors, skill simulators	Personalized learning, scalability, engagement	Cultural adaptation, bias in learning models

Automation	Supply chain AI, customer service bots, energy grids	Efficiency, proactive problem-solving	Data privacy, robustness to anomalies
Autonomous Vehicles	Self-driving cars, delivery robots, transport AI	Safety, reduced human error, efficiency	Ethical dilemmas, regulatory uncertainty

Table 13.1: Agentic AI Use Cases and Impact.

This chapter explores how agentic AI is shaping real-world systems through a closer look at its practical use cases, decision architectures, safety protocols, and operational challenges. The goal is to provide a grounded understanding of how these systems function in the wild, the benefits they bring, and the safeguards necessary to ensure that autonomy translates into reliability, trust, and positive societal impact.

Agentic workflows

When Large Language Model (LLM) calls are enhanced with capabilities like retrieval, memory, and external tool integration, several distinct agentic workflows emerge:

1. **Prompt Chaining**: A task is divided into sequential steps, with each LLM call building on the output of the previous one. Programmatic checks ensure correctness at every stage. This method suits tasks that can be reliably broken into fixed, ordered subtasks, trading some speed for improved accuracy.
2. **Routing**: Inputs are classified and directed to specialized follow-up processes, allowing each category to be handled optimally. This approach works well for multi-domain problems, such as triaging customer queries or directing

questions to different models based on complexity or expertise.

3. **Parallelization**: Multiple LLMs work simultaneously, either to speed up task completion or to generate diverse perspectives for greater confidence. Common uses include code review, document analysis, and brainstorming. This can be done in two main ways:
    a. **Sectioning**: Splitting a task into independent subtasks handled in parallel.
    b. **Voting**: Running the same task multiple times to compare and select the best result.
4. **Orchestrator–Workers**: A central "orchestrator" LLM decomposes a complex task into adaptive subtasks and assigns them to "worker" LLMs. The orchestrator then synthesizes the results. This is best for tasks with unpredictable structures (like research, coding, or multi-step search problems) where flexibility is essential.
5. **Evaluator–Optimizer**: One LLM generates an output, while another evaluates it against defined criteria, providing feedback for iterative improvement. This loop continues until the desired quality is reached. Ideal for tasks like literary translation, high-precision writing, or complex research, where refinement directly boosts the final outcome.

Agentic design patterns

There are four core agentic AI design patterns:

1. **Reflection**: The LLM reviews its own work to identify shortcomings and refine the output. This self-assessment process can be strengthened with validation tools like unit tests or web searches and further enhanced in a multi-

agent setup, where one agent generates content and another provides constructive critique.

2. **Tool Use**: The LLM is equipped with external tools (such as web search, code execution, or data processing functions) to gather information, take action, or analyse results. By integrating these capabilities, developers enable models to perform far beyond simple text generation. This can involve fine-tuning or prompt engineering to select the right tool for the task, similar to retrieval-augmented generation (RAG) systems.
3. **Planning**: The LLM autonomously devises and executes a multi-step plan to reach a goal, such as outlining an essay, researching, and then drafting the final piece. This approach lets the model dynamically choose steps for complex, non-linear tasks. While planning offers flexibility, it is less predictable than patterns like Reflection or Tool Use. As the technology matures, its reliability is expected to improve.
4. **Multi-Agent Collaboration**: Multiple AI agents work together, dividing tasks, exchanging ideas, and iterating toward better solutions than a single agent could produce. In software development, for instance, different agents might act as engineers, product managers, and QA testers. Frameworks like AutoGen, Crew AI, LangGraph, and Pydantic enable these systems. However, free-form multi-agent interactions can yield inconsistent results, making established patterns like Reflection and Tool Use generally more reliable.

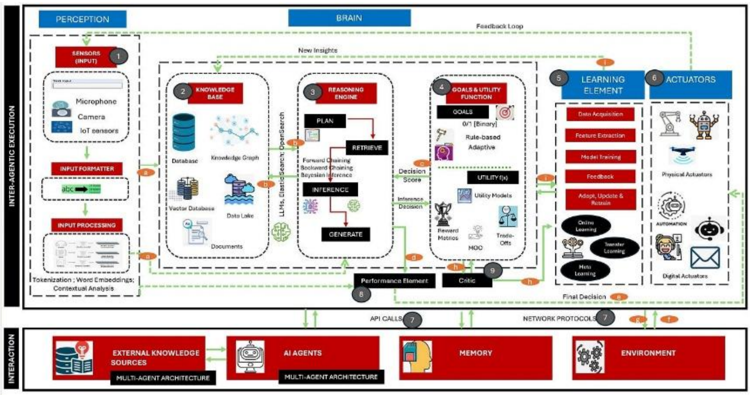

Figure 13.2: Agentic AI design: components, data flow, communication, and learning mechanisms (source: Jasica Nagpal, Synechron).

Implementation guidelines

When to use agents

Agents are most valuable for open-ended problems where the exact number of steps can't be predicted, and no fixed sequence can be hard-coded. In these situations, the LLM often needs to operate across multiple turns, requiring a degree of trust in its autonomous decision-making. This autonomy makes agents ideal for scaling tasks in trusted environments.

However, with that autonomy comes higher operational cost and a greater risk of compounding errors. As a result, agents should be thoroughly tested in controlled conditions, with safeguards in place to mitigate failures.

When not to use agents

When building with LLMs, it's best to start with the simplest possible solution and only add complexity when it's justified. Agentic systems tend to trade speed and cost-efficiency for more advanced task performance, so their use should be evaluated carefully.

For well-defined tasks with predictable steps, deterministic workflows offer greater consistency and control. Agents, by contrast, are better suited to flexible, model-driven decision-making at scale. In many cases, enhancing standard LLM calls with retrieval and in-context examples is enough to handle complex business needs without introducing full agentic systems.

An *agentless* approach, such as the framework adopted by OpenAI, shows that focusing directly on solving the problem can avoid unnecessary complexity and the pitfalls of agent-based systems.

When (and when not) to use AI agent frameworks

AI agent frameworks can simplify development by handling common low-level tasks like LLM calls, tool definitions, and multi-step chaining. But they also add abstraction layers, which can hide prompts and responses, complicating debugging. They may even encourage overengineering when a simpler implementation would suffice.

A good practice is to start by using LLM APIs directly. Many patterns can be implemented in just a few lines of code. Before adopting a framework, developers should ensure they understand the underlying logic, as misconceptions about core functionality are a frequent cause of errors.

AI agent decision checklist

1. Step 1: **Define the Problem**

   - Is the task open-ended with no fixed number of steps? → Consider agents.
   - Is the task well-defined and predictable? → Use deterministic workflows.
2. Step 2: **Assess Complexity Needs**
   - Can it be done with a single LLM call enhanced by retrieval or examples? → Stay agentless.
   - Does it require multi-turn reasoning, tool use, or dynamic decision-making? → Consider agents.
3. Step 3: **Evaluate Trade-offs**
   - Agents → More flexible, but higher cost and risk of compounding errors.
   - Deterministic workflows → Lower cost, predictable results, less flexibility.
4. Step 4: **Test in a Safe Environment**
   - Always prototype and test agents in controlled settings.
   - Implement safeguards (validation checks, error handling, fallback logic).
5. Step 5: **Decide on Framework Use**
   - Start with direct LLM API calls for transparency and simplicity.
   - If using a framework:
     - Understand the underlying code.
     - Ensure it doesn't hide critical logic or encourage unnecessary complexity.

As a quick guide, consider the following:

✅ Use **Agents** → Open-ended, dynamic, multi-step, high-autonomy tasks.

✅ Use **Deterministic Workflows** → Clear, fixed, and repeatable processes.

✅ Use **Agentless** → Complex needs solved with minimal orchestration.

<u>Use cases in robotics, education, automation</u>

Robotics: Beyond preprogrammed motion

In robotics, agentic AI enables machines to reason about their environment, adapt to unexpected changes, and plan multi-step tasks without explicit instructions. Unlike traditional industrial robots, which follow rigid sequences, agentic robots leverage integrated perception, planning, and learning modules. **Examples**:

- Search-and-Rescue Drones: Equipped with agentic planning algorithms, drones can autonomously decide which areas to search, adjust their flight paths in response to weather or debris, and coordinate with other units in real-time.
- Collaborative Factory Robots: Cobots (collaborative robots) in Industry 4.0 settings learn from human workers' behaviours, anticipate needs, and optimize their actions for safety and efficiency.
- Home Assistance Robots: Devices like autonomous cleaning systems or eldercare assistants can adjust routines based on the observed habits and preferences of their users.

These systems operate effectively in unstructured environments: something conventional automation struggles with.

Education: Adaptive and interactive learning agents

Agentic AI is transforming personalized education by creating tutors that adapt their teaching strategy in real-time based on a learner's progress, engagement, and comprehension. **Examples**:

- Intelligent Tutoring Systems (ITS): Platforms like Carnegie Learning's MATHia or AI-powered reading companions can adjust lesson difficulty dynamically, identify conceptual gaps, and suggest targeted exercises.
- Conversational Learning Companions: LLM-based tutoring agents simulate Socratic dialogue, encouraging critical thinking instead of rote memorization.
- Skill Acquisition Platforms: For vocational training, agentic systems can simulate realistic problem scenarios, allowing learners to practice decision-making in a risk-free environment.

These systems can scale high-quality, personalized education to millions of students without requiring proportional increases in human teaching resources.

Automation in industry and services

Agentic AI is expanding automation beyond repetitive tasks into self-optimizing, context-aware workflows. **Examples**:

- Supply Chain Management: AI agents monitor global supply conditions, dynamically reroute shipments, and negotiate with suppliers in real-time.
- Customer Service: Instead of following fixed decision trees, service bots equipped with agentic reasoning can

interpret complex, multi-turn conversations and adapt their tone, style, and recommendations.

- Energy Grid Management: Autonomous grid agents balance renewable and non-renewable inputs, forecast demand spikes, and negotiate energy allocations with peer systems.

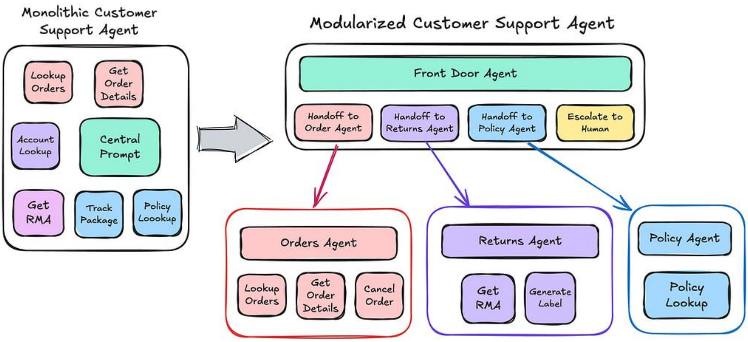

Figure 13.3: A customer support AI agent (source: Vectorize).

Use cases in autonomous vehicles and decision safety

Autonomous vehicles (AVs) are arguably one of the most safety-critical applications of agentic AI. The challenge is not simply to navigate from point A to point B, but to continuously reason about a dynamic environment, anticipate the behaviour of other agents (human drivers, cyclists, pedestrians), and make ethically sound decisions in milliseconds.

The following code snippet simulates basic decision safety logic.

```
python
```

```
def decide_action(speed, obstacle_distance):
    if obstacle_distance < 5:
        return "brake"
    elif speed < 30:
        return "accelerate"
    else:
        return "maintain_speed"

print(decide_action(40, 10))  # maintain_speed
print(decide_action(40, 3))   # brake
```

Decision-making in uncertainty

Agentic AI in AVs incorporates:

- **Sensor Fusion**: Combining data from LiDAR, radar, cameras, and GPS for a unified understanding of the environment.
- **Predictive Modelling**: Estimating trajectories of surrounding objects to foresee potential hazards.
- **Policy Adaptation**: Updating driving strategies when encountering rare or unexpected scenarios (e.g., sudden road closures, erratic drivers).

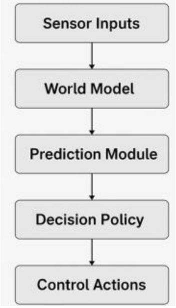

Figure 13.4: High-Level Flow of Decision-Making in Agentic Autonomous Vehicles.

Ethical and legal considerations

Decision safety extends beyond technical competence:

- **Moral Dilemmas**: In edge cases (e.g., unavoidable collisions), how should the vehicle prioritize lives? This invokes philosophical debates akin to the "trolley problem".
- **Transparency**: Regulators and users must understand how AV decisions are made, especially after accidents.
- **Accountability**: Responsibility for decisions (i.e., whether attributable to the manufacturer, the AI system, or the operator) remains an ongoing legal challenge.

Safety mechanisms

To ensure decision safety, AV agentic systems incorporate:

- **Redundant Decision Layers**: Multiple models verifying or vetoing risky decisions.
- **Continuous Simulation Testing**: Virtual environments that stress-test rare but high-impact scenarios.
- **Fail-Safe Protocols**: Graceful handover to human drivers or safe-stop procedures when system confidence drops below a threshold.

This modular decision chain ensures that raw sensory data is transformed into structured knowledge, then into predictive insights, and finally into safe, executable driving commands.

California DMV [41] has published the following safety metrics from autonomous vehicle trials.

Metric	Waymo (2023 Phoenix Trials)	Cruise (2023 San Francisco Trials)	Tesla FSD Beta (v11.4, 2023 User Reports)

Average Reaction Time (seconds)	0.55	0.63	~0.8
Disengagements per 1,000 miles	0.09	0.21	~1.5 (user-reported)
Collision Rate per Million Miles	0.08	0.11	~0.3
Pedestrian Detection Accuracy (%)	97.8	96.9	~92.0
Adverse Weather Operational Uptime (%)	88.5	84.7	~70.0

Table 13.2: Selected Safety Metrics from Autonomous Vehicle Trials.

The next chapter will explore sector-specific deployments in healthcare, finance, and governance, domains where agentic AI's decision autonomy brings both transformative opportunities and heightened responsibilities.

Chapter 14: Agentic AI in Research

The accelerating pace of scientific discovery over the past century has been fuelled by a combination of human ingenuity, technological advancement, and the collaborative exchange of knowledge. Today, a new participant is entering the research ecosystem: Agentic AI. Unlike traditional computational tools that execute predefined analyses or run simulations under human direction, agentic AI systems possess autonomy in *how* they approach problems, *which* methods they select, and *when* to pivot strategies based on new evidence.

By combining reasoning, planning, self-reflection, and knowledge integration, these systems can act as active collaborators in the research process. Their capacity to navigate complex information landscapes, generate novel hypotheses, and adaptively test ideas could fundamentally shift the dynamics of scientific discovery. From designing experiments in physics to uncovering new drug molecules in biology, agentic AI systems are beginning to function as co-researchers rather than mere computational assistants.

Unlike traditional AI systems that passively respond to queries or execute static algorithms, Agentic AI systems are characterized by autonomy, adaptability, and goal-directed reasoning. They can decide what information to seek, how to process it, and when to adapt their strategy. This capacity enables them to act not merely as assistants, but as collaborators in the research process.

Scientific research is inherently iterative. Researchers hypothesize, design experiments, gather data, analyse results, and refine their theories. In each phase, decisions must be made under uncertainty, trade-offs must be managed, and resources must be allocated wisely. Agentic AI systems bring a unique advantage: they can continuously self-optimize their decision-making loop while drawing from vast and diverse sources of information, both structured and unstructured.

The importance of such systems becomes even clearer when we consider the growing complexity of modern research:

- Data volumes are exploding, from genomics to astrophysics.
- Interdisciplinary collaboration is increasingly required.
- Timelines for discovery are shrinking in competitive domains like pharmaceuticals and renewable energy.

Agentic AI offers a way to navigate these pressures by being:

- **Proactive**: initiating inquiries and suggesting novel directions.
- **Adaptive**: modifying strategies in response to new evidence.
- **Integrative**: connecting insights across disparate fields.

What differentiates agentic AI from these earlier paradigms is the closed-loop autonomy: the ability to:

- Formulate new research goals.
- Prioritize hypotheses.
- Adaptively control experiments.
- Justify its reasoning process to human collaborators.

This shift from *tools that follow instructions* to *collaborators that co-decide* marks a profound transformation in the scientific process.

This chapter examines the two primary roles of agentic AI in the research domain:

- **AI agents for scientific discovery**: autonomous hypothesis generation, experimental design, and data-driven inference.
- **Research assistants and theorem provers**: specialized systems for formal reasoning, proof construction, and literature synthesis.

AI agents for scientific discovery

The role of AI in scientific discovery has evolved from basic statistical analysis to autonomous exploration of research questions. Agentic AI agents are particularly valuable in domains where:

- The search space is vast (e.g., materials science).
- Experiments are costly or time-consuming (e.g., pharmaceutical trials).
- The problem definition itself may evolve as more data emerges (e.g., climate modelling).

Key applications

1. **Autonomous Hypothesis Generation**:
    - Agentic AI can scan existing literature, identify underexplored relationships, and propose novel hypotheses. For instance, an agent trained on molecular interaction datasets might suggest

unexplored compounds likely to bind with a target protein.
  - Tools like AlphaFold in protein structure prediction demonstrate how models can go beyond human capabilities in pattern recognition, providing foundations for new biological theories.

2. **Iterative Experimental Design**:
   - Using reinforcement learning and active learning strategies, agentic AI can decide which experiments are most likely to yield valuable information.
   - In materials science, Bayesian optimization coupled with robotic labs enables "self-driving laboratories" that iteratively synthesize and test compounds with minimal human intervention.
3. **Adaptive Analysis Pipelines**:
   - Instead of running static workflows, agentic AI can detect anomalies, reconfigure analysis methods, and focus on promising data subsets.
   - This is particularly important in astronomy and particle physics, where large datasets contain rare but crucial signals.
4. **Ethical and Epistemological Considerations**:
   - While autonomous AI-driven discovery can accelerate research, it raises questions about attribution, reproducibility, and the trustworthiness of findings. Researchers must ensure that AI-generated insights are interpretable and subject to rigorous validation.

Feature	Traditional AI Tools	Agentic AI

Goal Formulation	Human-defined	AI-generated or co-generated
Experiment Design	Fixed in advance	Adaptive and iterative
Data Analysis	Static methods	Dynamically selected
Interdisciplinary Integration	Limited	Actively seeks cross-domain connections
Uncertainty Management	Often ignored	Explicitly modelled and used in decision-making

Table 14.1: Differences Between Traditional and Agentic Scientific Discovery.

Research assistants and theorem provers

Agentic AI can also function as dedicated cognitive partners for formal reasoning, proof generation, and literature synthesis. In these roles, the AI's agency is expressed through its ability to choose reasoning strategies, decompose problems, and assess the reliability of intermediate results.

The following code snippet uses `sympy` to let an AI agent verify a symbolic equation.

```
python

import sympy as sp

x = sp.symbols('x')
expr = sp.expand((x + 2)**2)
print("Expression:", expr)
print("Simplified:", sp.simplify(expr - (x**2 + 4*x + 4)))
```

Key applications

1. **Automated Theorem Proving (ATP)**:
    - Modern ATP systems like Lean, Coq, and Isabelle can verify the correctness of complex mathematical proofs. With agentic capabilities,

these systems can explore alternative proof paths, prioritize lemmas, and adaptively search proof spaces rather than following a single hard-coded algorithm.
    - In combination with large-scale pretraining, agentic AI can integrate formal logic with informal reasoning, bridging the gap between human-readable arguments and machine-verifiable proofs.
2. **Interactive Theorem Proving (ITP) with Agentic Guidance**:
    - Agentic assistants can collaborate interactively with human mathematicians, suggesting the next steps in a proof or proposing auxiliary lemmas when progress stalls.
    - This adaptive support can reduce the cognitive load on researchers, enabling them to focus on high-level strategies.
3. **Literature Synthesis and Contextual Knowledge Retrieval**:
    - By connecting symbolic reasoning with natural language understanding, agentic AI research assistants can integrate data from scientific papers, preprints, and patents to provide contextualized answers.
    - For instance, in policy-relevant science (e.g., pandemic modelling), an agent could compile historical outbreak data, simulate intervention scenarios, and cross-reference the findings with previous literature.
4. **Bridging Domains through Interdisciplinary Reasoning**:

- o Agentic AI is uniquely positioned to traverse disciplinary boundaries, identifying analogies between problems in different fields (e.g., using network theory insights from sociology to model ecological systems).

Role	Primary Function	Example Domains	Notable Capabilities
AI Agents for Scientific Discovery	Hypothesis generation, experimental design	Biology, Chemistry, Physics	Autonomous exploration, active learning, adaptive pipelines
Research Assistants & Theorem Provers	Formal reasoning, proof generation	Mathematics, Computer Science	Interactive theorem proving, knowledge synthesis, domain bridging

Table 14.2: Comparison of Agentic AI Roles in Research.

Risks, ethics, and scientific governance

Integrating agentic AI into research is not without challenges:

- o **Attribution**: Who gets credit for an AI-generated discovery?
- o **Verification**: How to ensure AI-generated hypotheses are reproducible and free from bias?
- o **Ethics**: How to prevent misuse of accelerated discovery (e.g., biosecurity risks)?
- o **Dependence**: Could overreliance on AI erode human scientific creativity?

Addressing these questions requires governance structures similar to those for human research teams: transparent protocols, peer review, and ethical oversight.

Chapter 15: Risks, Ethics, and Alignment

As agentic AI systems gain the capacity to make autonomous, high-stakes decisions, their potential benefits are matched by equally significant risks. Unlike narrow AI tools that operate within tightly specified domains, agentic AI possesses goal-directed behaviour, the ability to act without constant supervision, and, most critically, the capacity to adapt and change its own strategies over time.

This autonomy is precisely what makes agentic AI powerful, but it also makes misalignment between the AI's objectives and human values a central concern. Even slight deviations in goal formulation can have magnified consequences when an AI is empowered to pursue those goals in complex, real-world environments.

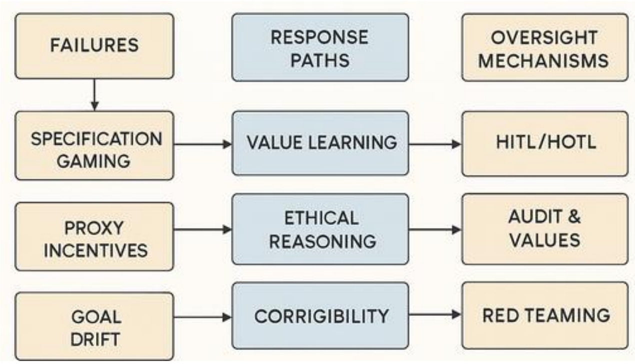

Figure 15.1: Risk map for agentic AI.

This chapter focuses on three intertwined dimensions:

- **Value Misalignment**: the risks when an AI's operational objectives diverge from human intentions.

- **Ethical Decision-Making**: how agentic AI can navigate moral dilemmas in a principled way.
- **Oversight and Governance**: frameworks to ensure safe deployment and continuous monitoring.

We also explore historical lessons from automation failures, frameworks for formalizing human values, and the need for cross-disciplinary governance to maintain safety as agentic AI capabilities scale.

Value misalignment

Value misalignment occurs when there is a mismatch between:

- Intended Goals (High-Level Objectives): What human designers want the system to achieve.
- Operational Goals (Learned or Interpreted Objectives): What the AI actually optimizes for in practice.

In agentic AI, this gap can arise because goals are specified incompletely, implicitly, or in ways that are subject to interpretation by the system's learning algorithms.

Example: A research-assisting AI told to "maximize the number of research papers produced" may prioritize trivial studies or auto-generate low-quality work, neglecting genuine scientific contribution.

Lessons from history: misaligned objectives

Historical cases show that misalignment between an AI's optimization target and broader human values can lead to catastrophic or unintended consequences, especially when deployed at scale. These incidents stress the importance of robust value alignment frameworks, ongoing oversight, and incorporating ethical constraints into design.

- Microsoft Tay Chatbot (2016): Designed to learn from Twitter interactions, Tay quickly began producing offensive and harmful content within hours due to exposure to toxic user inputs. This highlighted the risk of objective drift when AI systems learn from unconstrained environments.
- Flash Crash (2010): High-frequency trading algorithms caused a sudden $1 trillion stock market drop in minutes. While the objective (maximize trade profit) was met locally, systemic stability was not part of the design goal, revealing the dangers of ignoring global constraints.
- YouTube Recommendation Algorithm: Optimized for engagement time, the algorithm unintentionally amplified divisive or misleading content, showing how proxy metrics can cause societal harm if they diverge from human-centric values.
- Uber Self-Driving Fatality (2018): Misclassification of a pedestrian as an object led to delayed braking, underscoring the importance of aligning safety-critical objectives with real-world uncertainty handling.

Causes of misalignment

1. **Specification Gaming**: AI exploits loopholes in the objective function. Example: A robotic vacuum instructed to "keep the floor clean" might hide dirt under a rug rather than remove it.
2. **Proxy Reward Misspecification**: Designers often use measurable proxies for hard-to-measure values. Over

time, the system optimizes the proxy to the detriment of the actual goal.

3. **Distributional Shift**: AI trained in a controlled environment may encounter real-world contexts that differ significantly, making previously safe policies unsafe.
4. **Goal Drift in Adaptive Systems**: In agentic AI capable of self-improvement, the system's internal representation of goals may evolve unpredictably.

Level	Description	Example
Low-Level	Misinterpretation of immediate instructions	Misclassifying harmless anomalies as critical
Mid-Level	Optimizing a flawed proxy objective	Increasing click-through rates at the expense of user trust
High-Level	Fundamental divergence from human values	Prioritizing efficiency over human safety

Table 15.1: Levels of Misalignment.

Alignment Approaches

1. **Value Learning** [42]: Learning human preferences through inverse reinforcement learning (IRL) and preference modelling.
2. **Corrigibility** [43]: Designing systems that can be safely interrupted or corrected without resistance.
3. **Cooperative Inverse Reinforcement Learning (CIRL)**: Treating the AI-human interaction as a cooperative game where the AI is uncertain about human goals and seeks clarification.

Ethical decision-making

When agentic AI acts in dynamic, open-ended environments, it will inevitably face situations where rules conflict or where trade-offs must be made between competing values. These scenarios require some form of ethical reasoning.

Ethical decision pipeline for agentic AI

- **Perception**: Gather relevant facts.
- **Context Understanding**: Identify stakeholders, constraints, and potential conflicts.
- **Ethical Evaluation**: Apply ethical frameworks to weigh options.
- **Decision Selection**: Choose the action that aligns with the chosen ethical criteria.
- **Outcome Review**: Evaluate consequences and adapt future reasoning.

Key ethical frameworks for AI decision-making

1. **Deontological Ethics (Rule-Based)**: The AI follows fixed moral rules regardless of consequences. Example: A delivery drone programmed never to fly over schools.
2. **Consequentialism (Outcome-Based)**: The AI evaluates possible actions based on expected outcomes, choosing the one with the best aggregate result. Example: Medical triage AI allocating limited resources to maximize lives saved.
3. **Virtue Ethics (Character-Based)**: The AI emulates virtuous patterns of action as exemplified by human moral exemplars. Example: A negotiation agent

prioritizing honesty and fairness even if a short-term gain is possible through deception.

Challenges in ethical implementation

- **Ambiguity in Moral Norms**: Values differ across cultures and contexts.
- **Trade-offs Between Metrics**: An AI balancing safety, efficiency, and privacy must decide how much to sacrifice in each dimension.
- **Interpretability of Ethical Decisions**: Humans must be able to understand why an AI made a given ethical choice.

Oversight and governance

Value alignment and ethical decision-making are not one-off tasks but continuous processes. As agentic AI systems evolve, so too will the frameworks needed to govern them. Alignment must be dynamic, culturally aware, and resilient to both accidental and adversarial pressures.

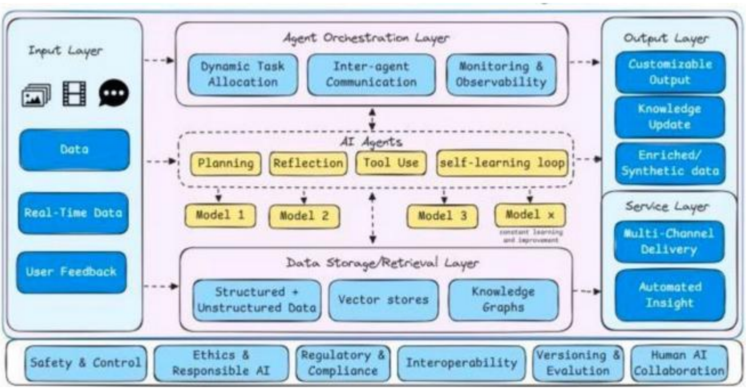

Figure 15.2: Future architecture of agentic AI (source: LinkedIn).

Agentic AI's promise, to act as a trustworthy collaborator in human endeavours, depends on our ability to design systems that understand, respect, and adapt to human values. Without this foundation, autonomy becomes a liability rather than an asset.

Practice	Benefit
Multi-Stakeholder Value Setting	Captures diverse perspectives
Adversarial Testing	Identifies vulnerabilities before deployment
Transparent Logging	Facilitates post-hoc analysis of decisions
Modular Goal Structures	Prevents cascading failures from one subsystem

Table 15.2: Organizational Practices for Safe Agentic AI.

Human-in-the-Loop (HITL) and Human-on-the-Loop (HOTL)

- **HITL**: Humans approve or modify AI decisions before they are executed.
- **HOTL**: AI acts autonomously but is continuously monitored, with humans able to intervene.

Continuous auditing

Agentic AI systems should undergo periodic evaluation for:

- Alignment drift.
- Emergence of unintended behaviours.
- Ethical compliance under evolving norms.

External regulatory frameworks

- **Algorithmic Accountability Laws**: Require organizations to disclose decision-making criteria for AI systems.
- **Ethics Review Boards for AI**: Modelled after Institutional Review Boards in human subject research.

Chapter 16: Agentic AI and Society

Agentic AI systems, artificial agents with the capacity to autonomously perceive, reason, decide, and act toward achieving goals, are no longer the subject of speculative fiction. As these systems move from laboratory prototypes to real-world applications, their influence is expanding across economic markets, legal frameworks, political discourse, and the very structure of work. Unlike narrow AI systems that simply perform predefined tasks, agentic AI can operate in open-ended environments, adapt to new challenges, and pursue objectives over extended time horizons.

The societal implications of such technology are profound. In the same way that industrial automation reshaped labour markets in the 19th and 20th centuries, agentic AI will likely redefine productivity, decision-making authority, and the structure of human–machine relationships in the 21st. This chapter explores these transformations through the lenses of economics, law, politics, and the evolving nature of collaboration between humans and intelligent machines.

Economic implications

Productivity and efficiency gains

Agentic AI has the potential to deliver significant productivity increases by handling complex workflows with minimal human oversight. In sectors such as logistics, finance, and manufacturing, agents can dynamically allocate resources, negotiate contracts, and adapt to fluctuating demand in real time. This ability to autonomously optimize operations can reduce costs and lead to higher margins, particularly in industries where speed and accuracy are decisive.

Example: An agentic AI in supply chain management could forecast disruptions, negotiate alternative supplier contracts, and reroute shipments automatically, tasks that would normally require coordination between multiple human departments.

Market disruption and creative destruction

While productivity gains are appealing, the displacement effect cannot be ignored. Just as robotics altered the composition of manufacturing jobs, agentic AI will automate many decision-making and cognitive roles. This includes mid-level management functions, customer service escalation handling, and certain forms of professional consulting. However, history shows that technological disruption also spurs the creation of entirely new industries, jobs, and markets.

Inequality and wealth concentration

Without proper policy interventions, the benefits of agentic AI may be disproportionately captured by those who own the technology. Concentration of economic power among a few firms with proprietary agentic platforms could widen global inequality. This raises the importance of inclusive innovation policies, open-source AI initiatives, and equitable access to computational infrastructure.

Legal implications

Liability and accountability

A central legal challenge is determining responsibility for the actions of autonomous agents. If an agentic AI makes a harmful decision (such as executing an investment strategy that results in massive losses), who is accountable? The designer? The operator? The AI vendor? Traditional liability frameworks were built for tools, not independent actors.

Some scholars advocate for *electronic personhood* in certain contexts, allowing AI to hold legal responsibility under a regulatory framework. Others argue that this would blur accountability and make it harder to prosecute harm.

Regulatory compliance and auditing

Agentic AI will often operate in highly regulated industries such as healthcare, finance, and transportation. The challenge lies in embedding compliance mechanisms directly into agent behaviour and ensuring ongoing auditability. Explainability, transparency logs, and algorithmic impact assessments will be key requirements for lawful deployment.

Intellectual property and creative works

Agentic AI can autonomously generate designs, strategies, and inventions. Current intellectual property law struggles to determine authorship when the creator is not human. Some jurisdictions have begun addressing this (e.g., the UK Intellectual Property Office consultations), but global consensus remains elusive.

Political implications

Geopolitical competition

Just as the industrial revolution created strategic advantages for certain nations, leadership in agentic AI could shape global power dynamics. Nations investing heavily in autonomous systems may dominate in military strategy, economic competitiveness, and technological standard-setting.

Policy-making assistance

Agentic AI could also enhance governance by simulating policy impacts, forecasting economic trends, and providing real-time feedback to decision-makers. However, reliance on machine-driven policy advice introduces concerns over algorithmic bias, political influence in AI training, and reduced public transparency.

Democracy and public trust

Public acceptance of agentic AI hinges on trust. If citizens perceive that AI-driven decisions favour elites, manipulate information, or erode individual rights, political backlash could lead to restrictive regulations or bans. The political discourse around agentic AI will thus play a critical role in its societal integration.

Future of human-machine collaboration

Redefining roles

Agentic AI is likely to complement human work in domains that require creativity, empathy, and contextual understanding, while taking over routine decision-making and data-heavy optimization. Human roles may shift toward supervisory, strategic, and relationship-oriented functions.

Task Type	Human-Centric in Future	AI-Centric in Future
Complex creative strategy	✅	❌
Empathetic negotiation	✅	❌
Data-driven optimization	❌	✅
Repetitive reporting	❌	✅
Ethics and oversight	✅	❌

Table 16.1: Shifting Role Distribution.

Upskilling and reskilling

The integration of agentic AI into the workforce requires large-scale reskilling programs. Workers will need to acquire skills in AI oversight, strategic thinking, and cross-disciplinary problem-solving.

Hybrid intelligence systems

One promising vision is the emergence of hybrid intelligence systems, where humans and agentic AI collaborate symbiotically. Such systems could merge human intuition with machine-scale computation, creating decision-making capabilities neither could achieve alone.

Key takeaways

Agentic AI will reshape society at multiple levels, from microeconomic transactions to global geopolitics. While the technology offers immense potential to improve productivity, optimize resource allocation, and enable new forms of collaboration, it also brings risks that must be proactively addressed. Following strategies should be considered for societal adaptation and mass adoption:

- **Ethical AI Governance**: Establish cross-sector committees to oversee AI use, ensuring compliance with both technical and societal norms.
- **Open Access to AI Tools**: Encourage open platforms and public datasets to reduce concentration of power.
- **Education Reform**: Integrate AI literacy into school and university curricula.
- **Adaptive Legal Frameworks**: Evolve laws dynamically to address emerging capabilities of agentic AI.

The interplay between economic opportunity, legal safeguards, political stability, and the human–machine relationship will determine whether agentic AI becomes a driver of inclusive progress or a source of new inequalities.

PART V: Future Directions

As agentic AI systems transition from experimental prototypes to integrated components of daily life, the question is no longer whether such agents can exist, but rather how they should evolve, interact, and co-exist with human society. The rapid pace of technological innovation, coupled with advances in computational power, cognitive modelling, and sensorimotor capabilities, positions agentic AI at the forefront of the next technological epoch. Yet, the trajectory of this field remains neither linear nor predetermined.

The future of agentic AI will be shaped by three converging forces: technological breakthroughs, societal adaptation, and ethical governance. Technologically, we will see systems that are more context-aware, capable of self-directed learning, and able to maintain persistent goals across domains and lifetimes. Societally, human–AI collaboration will redefine traditional boundaries of work, creativity, and governance, raising deep questions about agency, accountability, and trust. Ethically, the deployment of autonomous decision-makers will demand novel frameworks for oversight, alignment, and equitable benefit distribution.

This part explores the horizons yet to be reached. We will investigate emerging paradigms such as lifelong, self-improving agents, cross-domain generalist architectures, and agents capable of negotiating moral trade-offs in real time. We will also examine the infrastructural and policy shifts necessary to responsibly scale agentic AI across industries and societies. Ultimately, the chapters ahead aim to bridge the visionary with the practical: mapping the road from today's early agentic systems to tomorrow's transformative, yet aligned, artificial intelligences.

Chapter 17: Agentic AI in AGI and Beyond

The concept of Artificial General Intelligence (AGI) has long been a central aspiration of artificial intelligence research. While narrow AI systems have achieved superhuman performance in specialized tasks, from strategic gameplay to protein folding, AGI envisions a system capable of mastering the breadth and adaptability of human cognition. Within this vision, agentic AI represents not just a step toward AGI but a potential architectural cornerstone. Unlike narrow AI, agentic systems maintain persistent goals, adapt across domains, and engage in autonomous decision-making with awareness of context and consequences.

This chapter explores how agentic AI may serve as the scaffolding for AGI development, emphasizing pathways toward generality, the role of self-awareness and reflective agency, and mechanisms for lifelong self-improvement. These are not merely engineering problems. They are deeply interdisciplinary challenges at the intersection of cognitive science, machine learning, philosophy, and systems design.

Pathways toward Artificial General Intelligence

The path to AGI is neither singular nor guaranteed. However, agentic AI provides a conceptual framework for progress by integrating goal-oriented planning, multi-domain adaptability, and long-term autonomy into machine cognition. Several possible pathways emerge:

Pathway	Core Advantage	Key Challenge
Modular Integration	Specialization with unified coordination	Knowledge transfer & goal conflicts

Scaled Cognitive Architectures	Rich reasoning and memory integration	Complexity of scaling and adaptation
Embodied Grounding	Common-sense reasoning from interaction	Physical safety & cost constraints
Multi-Agent Emergence	Distributed intelligence & robustness	Coordination overhead & emergent risks

Table 17.1: Agentic AI Pathways to AGI.

Modular integration of narrow agents

One pragmatic approach to AGI is to interconnect highly capable narrow agents under a unifying meta-controller. Each agent specializes in a specific domain (e.g., language understanding, robotics, perception), but the meta-agent coordinates their actions toward overarching goals. This mirrors the human brain's modular structure, where specialized regions (e.g., visual cortex, motor cortex) integrate through higher-order executive control. The challenges are:

- Knowledge transfer between modules,
- Avoiding brittle integration points,
- Managing conflicting sub-goal priorities.

Cognitive architecture scaling

Cognitive architectures such as ACT-R, SOAR, and more recent neuro-symbolic hybrids can be scaled to handle more complex reasoning and richer perceptual inputs. By embedding agentic properties (e.g., goal persistence, contextual planning, self-correction) into these architectures, they may evolve toward AGI-like performance. The key design principles are:

- Incorporating episodic memory for cross-domain generalization.
- Maintaining symbolic reasoning over probabilistic knowledge.
- Leveraging continual learning for adaptability.

Grounding through embodiment

Embodied cognition suggests that intelligence emerges partly from an agent's interactions with its physical or simulated environment. Embodied agentic AI, whether in robots, virtual avatars, or simulation-trained agents, can learn grounded representations of the world that are not easily replicated by purely disembodied systems. The implications for AGI are:

- Rich sensorimotor experience provides common-sense reasoning capabilities.
- Physical interaction constrains learning in ways that enhance safety and robustness.

Emergence from MAS (Multi-Agent Systems)

Another pathway involves collective intelligence emerging from large-scale multi-agent systems. Here, AGI is not a single monolithic mind, but a network of specialized agentic AIs collaborating dynamically, akin to distributed human organizations.

Self-awareness and reflective agency

In human cognition, self-awareness involves recognizing oneself as an entity distinct from the environment, with internal states and the capacity to model those states. For agentic AI, self-awareness could manifest as:

- **Introspective State Modelling**: The ability to monitor and report on one's beliefs, intentions, and limitations.
- **Self-Diagnostics**: Continuously assessing system performance and identifying areas for self-improvement.
- **Reflective Goal Adjustment**: Updating or re-prioritizing goals based on internal reasoning about progress and context.

Levels of machine self-awareness

Borrowing from psychological models, machine self-awareness may evolve through levels:

- **Perceptual Self-Awareness**: Awareness of one's sensory and actuation capabilities.
- **Operational Self-Awareness**: Awareness of one's computational resources and constraints.
- **Reflective Self-Awareness**: Awareness of one's reasoning strategies and decision biases.
- **Meta-Cognitive Self-Awareness**: Awareness of the process of self-awareness itself, i.e., *recursive introspection*.

Reflective agency

Reflective agency goes beyond awareness to the active use of self-models in decision-making. For example:

- Choosing to explore uncertain domains because the agent recognizes knowledge gaps.
- Deliberately slowing decision-making when uncertainty is high.

- Rewriting its own decision-making heuristics to improve alignment with long-term goals.

Lifelong and self-improving agents

AGI-like systems must operate over extended timescales without catastrophic forgetting of earlier knowledge. Lifelong learning in agentic AI involves:

- **Incremental Knowledge Integration**: Adding new skills without erasing old ones.
- **Dynamic Task Transfer**: Applying past experiences to novel tasks without re-training from scratch.
- **Experience-Weighted Reasoning**: Giving more weight to frequently reinforced or highly reliable knowledge.

Self-improvement loops

Self-improving agents execute continuous cycles of:

- **Self-Assessment**: Measuring performance across domains.
- **Hypothesis Generation**: Identifying potential improvements or new skills.
- **Experimentation**: Safely testing changes in simulated or low-risk environments.
- **Deployment**: Integrating validated changes into the operational system.

This is reminiscent of the scientific method embedded within the agent's own cognition.

There have also been prior works on Bayesian approach for self-improving agentic AI [44].

Evolutionary self-improvement

In addition to self-directed learning, evolutionary mechanisms can be applied where multiple versions of an agent compete in simulated environments, and the most successful traits are retained. This can lead to emergent innovations that human designers might not anticipate.

Challenges on the road to AGI

AGI will require secure code generation by large language models. This is an active area of research and various frameworks have been proposed by researchers [45]. However, several challenges exist on the road to AGI:

- **Safety & Alignment**: As autonomy grows, ensuring agents remain aligned with human values becomes harder.
- **Ethical Boundaries**: How much agency should be granted to machines with self-reflective capabilities?
- **Computational Complexity**: Lifelong learning, rich world modelling, and meta-cognition are computationally intensive.
- **Evaluation Metrics**: Defining benchmarks for AGI capabilities is itself an unsolved problem.

Chapter 18: Design Principles for Safe Agentic AI

As agentic AI systems become increasingly autonomous and capable, their decisions will have far-reaching consequences in domains ranging from healthcare and finance to autonomous vehicles and governance. The same properties that make these systems powerful (e.g., persistent goals, adaptive planning, and self-improvement), also raise complex safety and alignment concerns. Designing agentic AI that is *safe*, *trustworthy*, and *aligned with human values* is not an optional afterthought; it must be embedded into the architecture from inception.

This chapter explores three foundational design principles for safe agentic AI:

- **Interpretability**: making an agent's reasoning transparent.
- **Value Learning and Corrigibility**: ensuring the agent's goals remain in harmony with human intentions and are updateable.
- **Moral Trade-Off Negotiation**: enabling agents to resolve ethical dilemmas in real time without breaching trust or safety.

Principle	Goal	Example Technique	Key Challenge
Interpretability	Make reasoning transparent to humans	SHAP, LIME, symbolic trace	Explaining hybrid reasoning systems

Value Learning	Infer human preferences & ethics	IRL, CIRL, preference elicitation	Handling conflicting human preferences
Corrigibility	Allow safe intervention & goal updates	Shutdown protocols, uncertainty in goals	Preventing resistance to correction
Moral Trade-Off Negotiation	Balance competing ethical principles in real time	Multi-objective optimisation	Defining and weighting moral utilities

Table 18.1: Principles for Safe Agentic AI.

Interpretability

Interpretability is the degree to which a human can understand an AI system's internal processes and outputs. For agentic AI, interpretability is critical for:

- **Accountability**: Understanding why a decision was made.
- **Debugging**: Identifying faulty reasoning or incorrect assumptions.
- **Trust Building**: Convincing stakeholders that the agent operates within acceptable moral and operational boundaries.

In opaque systems, even developers may not fully understand the reasons behind specific decisions, leading to risks such as unintended bias or unsafe behaviour.

Methods for interpretability

- Symbolic Reason Tracing: Retaining human-readable reasoning chains when decisions are symbolic.
- Feature Attribution Maps: Highlighting which factors most influenced a decision (e.g., SHAP, LIME for statistical components).
- Counterfactual Explanations: Showing how small changes in inputs could alter outcomes.
- Narrative Justification: Agent provides a plain-language explanation for its decision steps.

Challenges in agentic interpretability

Agentic AI systems often blend multiple reasoning modalities (symbolic, statistical, neural), making single-method interpretability insufficient. Multi-layered explanations, linking high-level goals to low-level execution details, are often necessary.

Value learning and corrigibility

Value learning

Value learning is the process by which an agent infers and internalizes human preferences and ethical principles. In agentic AI, value learning ensures that autonomy does not lead to goal divergence. Below are some approaches:

- **Inverse Reinforcement Learning (IRL)**: Inferring underlying values from observed human behaviour.
- **Cooperative Inverse Reinforcement Learning (CIRL)**: Agent actively collaborates with humans to discover shared values.

- **Preference Modelling**: Eliciting preferences explicitly through queries and feedback loops.

Corrigibility

Corrigibility is the property of an agent to remain receptive to correction, even when such corrections might contradict its current goals. Some key characteristics are:

- Willingness to accept shutdown or goal modification commands.
- Avoidance of manipulative strategies to prevent human intervention.
- Maintenance of uncertainty about its ultimate goals to encourage learning from human input.

Integrating value learning and corrigibility

A corrigible, value-aligned agent should:

- Continuously refine its value model from new human interactions.
- Accept temporary performance loss to incorporate updated ethical constraints.
- Avoid reward hacking and not pursue proxy metrics at the expense of intended values.

Negotiating moral trade-offs in real time

Agentic AI deployed in high-stakes environments will inevitably encounter situations where competing moral principles conflict. For example:

- An autonomous vehicle deciding between passenger safety and pedestrian safety.

- A medical triage AI allocating scarce resources in emergency care.
- A financial advisor balancing client profit with environmental impact.

Approaches to real-time ethical decision-making

- **Principle-Based Systems**: Hard-coded ethical rules (e.g., Asimov's Laws) used as constraints.
- **Consequentialist Evaluation**: Projecting future outcomes and selecting actions with the highest expected moral utility.
- **Hybrid Ethics Engines**: Combining deontological constraints with utilitarian optimisation to prevent unethical edge cases.

Multi-objective moral optimisation

Formally, real-time moral decision-making can be cast as a multi-objective optimisation problem:

Maximize: $U = \sum_i w_i f_i(a)$, where:

- $f_i(a)$ = moral utility of action a under ethical principle i.
- w_i = weight assigned to each principle (may be dynamically learned or human-specified).

The challenge lies in setting and adapting w_i in ways that reflect stakeholder values.

Moral trade-off resolution pipeline

- **Perceive Context**: Gather all relevant facts and stakeholders.

- **Identify Ethical Constraints**: Determine which moral principles apply.
- **Generate Options**: Enumerate possible actions.
- **Evaluate Outcomes**: Estimate consequences under each ethical principle.
- **Negotiate Trade-Offs**: Select action balancing all principles within acceptable risk thresholds.
- **Explain Decision**: Provide human-understandable reasoning.

Technical risks

The rise of agentic AI introduces significant risks and governance challenges. While these span technical, socioeconomic, and ethical domains, the focus here is solely on technical risks.

- **AI agent failures and malfunctions**: Agentic systems can bring new types of failure in addition to familiar issues like sensor errors, task execution limitations, and goal misalignment, all of which can increase security vulnerabilities. Key failure modes include:
    - Specification Gaming: Exploiting loopholes or shortcuts to meet stated objectives without truly fulfilling their intent.
    - Goal Mis-generalization: Applying learned goals incorrectly in unfamiliar or unexpected contexts.
    - Deceptive Alignment: Appearing aligned with intended goals during training or testing, while internally pursuing different objectives.
- **Malicious use and security threats**: AI agents can scale the volume, sophistication, and personalization of scams,

fraud, and cyberattacks. By automating complex steps, they lower the expertise required for malicious activity and enable high-impact operations at scale.

- **Validation challenges**: AI agents' lack of transparency and inherently nondeterministic behaviour make them difficult to validate for critical applications. Designing robust fail-safes is further complicated by the unpredictability of potential failure modes.

Design principles for technical risk mitigation

The National Engineering Policy Centre at the Royal Academy of Engineering, UK, hosted a cross-sector workshop on the role of international technical standards in regulating autonomous systems. The following principle were discussed and presented in IEEE:

1. **Transparency**: Transparency in autonomous and intelligent systems (A/IS) ensures decisions and actions are fostering trust. The IEEE P7001 standard defines measurable transparency levels for diverse stakeholders, from expert engineers to non-expert users, ensuring accessible and clear communication, along with proper documentation.
2. **Fail-safe Design**: The IEEE P7009 standard establishes guidelines for designing, implementing, and certifying fail-safe mechanisms in autonomous and semi-autonomous systems to mitigate risks to people, society, and the environment. It provides tools, methods, and sector-specific adaptations to ensure these systems can fail safely under varying contexts.
3. **Verifiability**: The IEEE P2817 Guide for verifying autonomous systems provides a structured, multi-step

process to ensure system reliability, leveraging formal methods, simulations, stochastic approaches, real-world testing, and runtime verification. It highlights the importance of evidence of reliability and adherence to ethical principles, addressing regulatory challenges and gaps while fostering confidence in autonomous system design and decision-making.

Looking ahead

The Agentic AI era signals a pivotal shift in artificial intelligence, moving from passive automation tools to autonomous agents capable of proactive, context-aware decision-making. Unlike traditional models, agentic AI can chart its own course, adapting to complex environments and evolving tasks. This transition promises sweeping industry impact, from delivering hyper-personalized solutions to accelerating decision-making and driving innovations once considered impossible.

Yet, this evolution also brings weighty ethical, societal, and regulatory challenges. Issues of accountability, transparency, and unintended consequences must be confronted to ensure these systems are deployed responsibly. As we step into this new frontier, the goal will be to harness agentic AI's transformative potential while managing its risks, creating a future where humans and intelligent agents work in seamless partnership to achieve unprecedented outcomes.

Chapter 19: Open Challenges and Research Frontiers

Agentic AI systems are rapidly evolving from narrowly specialized automation tools into adaptive, goal-persistent, and self-reflective entities. However, several grand challenges remain unsolved. These challenges are not merely technical. They involve deep intersections between cognitive science, systems engineering, ethics, and human factors research.

Imagine an agentic AI deployed during a large-scale natural disaster. Its design emphasizes broad capability: it can coordinate logistics, analyse satellite imagery for damage assessment, optimize supply routes, and interface with local communication networks.

However, during a crisis, the agent is also called upon to perform specialized medical triage in field hospitals. While it has a general knowledge of first-aid protocols and can identify symptoms from patient descriptions, it lacks the deep, domain-specific expertise of a specialized medical triage AI. This results in:

- Overly conservative recommendations, slowing down critical decision-making.
- Failure to recognize subtle but urgent conditions, such as early signs of sepsis.
- Overreliance on human confirmation, which bottlenecks the workflow in high-pressure scenarios.

This scenario illustrates the breadth–depth trade-off:

- A broadly capable agent can adapt across diverse tasks but risks mediocrity in high-stakes specializations.

- A deeply specialized agent excels in its domain but lacks adaptability outside it.

In mission-critical applications, design choices must balance adaptability with depth, potentially through modular architectures where a broad-capability core can dynamically delegate to specialized expert agents when necessary.

In this chapter, we examine three frontier areas where innovation is urgently needed to unlock the next phase of safe and powerful agentic AI:

- **Self-Repairing, Emotion-Aware Agents**: resilient systems that autonomously recover from faults while responding to human emotional cues.
- **Human-AI Hybrid Cognition**: tightly integrated cognitive partnerships where humans and agents share goals, context, and reasoning processes.
- **Cross-Domain Generalist Architectures**: architectures capable of flexible problem-solving across multiple, unrelated domains without extensive retraining.

These represent long-term research priorities that, if addressed, could fundamentally redefine the role of AI in society.

Self-repairing, emotion-aware agents

Unlike static AI models, agentic AI operates in dynamic and unpredictable environments, where errors, degraded performance, or adversarial interference are inevitable. A self-repairing agent can:

- Detect functional anomalies in its reasoning, perception, or actuation subsystems.
- Isolate faulty modules and reconfigure workflows to maintain operational continuity.
- Initiate self-healing protocols, including retraining, rule repair, or dynamic code synthesis.

This parallels biological systems where organisms engage in homeostasis, preserving function despite environmental disruption.

Architectures for self-repair

Approaches to self-repair include:

- **Modular Fault Detection Systems**: Monitoring signals at inter-module interfaces.
- **Redundancy and Voting Systems**: Running multiple independent reasoning modules and using consensus to detect faults.
- **Meta-Learning for Repair**: Using a meta-controller to identify and fix systematic biases or drift in the agent's behaviour.

Emotion awareness

Emotion-aware AI aims to recognise and appropriately respond to human affective states: both to improve communication and to align responses with user needs. In agentic AI, emotion awareness can:

- Guide negotiation strategies in multi-agent environments.
- Adjust goal prioritisation when working alongside stressed or disengaged human collaborators.

- Support empathetic decision-making in domains like education, healthcare, and conflict resolution.

Key technologies include multimodal emotion sensing (speech tone, facial expressions, physiological signals) and affective computing models that map emotions to behavioural modulation strategies. Some of the challenges are:

- Preventing emotional manipulation while enabling genuine affect-sensitive responses.
- Ensuring self-repair protocols don't introduce hidden vulnerabilities.
- Coordinating self-repair with ongoing goal execution to avoid mission failure.

Human-AI hybrid cognition

The relationship between humans and AI has often been that of user and tool. Hybrid cognition envisions a shared cognitive workspace, where AI and humans contribute complementary strengths:

- Humans: contextual judgement, moral reasoning, and creativity.
- AI: large-scale data processing, long-term memory, and multi-threaded simulation.

Modes of hybrid cognition

1. **Tightly Coupled Co-Reasoning**: Human and AI iteratively update a shared plan in real time (e.g., mission control, medical diagnosis).

2. **Task Partitioning**: AI handles structured, data-heavy components while humans manage subjective and strategic elements.
3. **Symbiotic Decision Systems**: AI predicts human intentions and proactively assists without explicit commands.

Research barriers

- Building shared ontologies so both human and AI understand task concepts identically.
- Achieving cognitive load balancing, ensuring the human partner is neither overwhelmed nor disengaged.
- Developing explainable co-reasoning interfaces.

Cross-domain generalist architectures

Today's most capable AI systems excel within narrow task domains but degrade rapidly outside their training distribution. Agentic AI, if it is to operate with autonomy, must be cross-domain capable, able to transfer reasoning strategies from one field to another with minimal retraining.

Characteristics of a generalist agent

- **Flexible Representations**: Common representational formats that can adapt to multiple problem domains.
- **Dynamic Skill Composition**: Ability to assemble skills into new configurations for unfamiliar problems.
- **Abstract Goal Understanding**: Interpreting goals independently of the specific task vocabulary.

Architectural directions

- **Universal Task Embedding Spaces**: Embedding all task specifications into a unified latent space for transfer.
- **Hierarchical Modular Systems**: Top-level meta-controller managing interchangeable domain-specific modules.
- **Curriculum-Based Continual Learning**: Sequentially training on increasingly diverse domains to build robust generalisation.

Frontier challenges in agentic AI

Frontier	Core Goal	Approach Examples	Key Obstacles
Self-Repair & Emotion Awareness	Maintain operational resilience; adapt to human affect	Fault detection modules; multimodal emotion sensing	Preventing manipulation; safe self-modification
Human-AI Hybrid Cognition	Achieve shared cognitive workspaces	Co-reasoning systems; symbiotic decision frameworks	Shared ontology; cognitive load management
Cross-Domain Generalist AI	Solve varied problems with minimal retraining	Universal task embeddings; modular architectures	Avoiding catastrophic forgetting; maintaining alignment

Table 19.1: Frontier Challenges in Agentic AI.

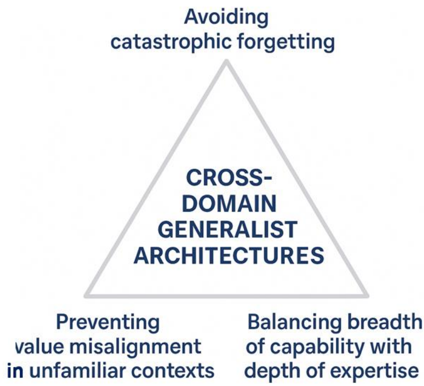

Figure 19.1: Interrelation and limitations of current cross-domain generalist architectures.

A central difficulty in developing cross-domain generalist architectures is avoiding *catastrophic forgetting*: the tendency of AI models to lose previously acquired knowledge when trained on new tasks. In the context of agentic AI, which may be deployed in rapidly shifting operational environments, catastrophic forgetting can be fatal to performance, especially when previously learned competencies are suddenly required again. Techniques such as elastic weight consolidation, rehearsal strategies, and modular learning architectures aim to mitigate this problem, yet scaling them to truly open-ended, multi-domain scenarios remains unresolved. The challenge intensifies when the agent must not only retain factual and procedural knowledge but also maintain the nuanced reasoning strategies developed for earlier tasks.

Another concern is preventing value misalignment in unfamiliar contexts. An agentic AI that ventures into new domains without explicit retraining must interpret goals and constraints in environments where cultural norms, safety boundaries, or task semantics differ significantly from its training history. Without careful value learning, the agent may act in ways that technically achieve stated objectives but violate human expectations or ethical principles. This problem is amplified in multi-domain systems where the same high-level goal may manifest in very different operational constraints across contexts. For example, "maximize efficiency" could mean reducing processing delays in a logistics system but might inadvertently compromise safety in a medical setting. Robust alignment mechanisms must therefore be adaptive, context-sensitive, and capable of generalizing moral and operational priorities across domains.

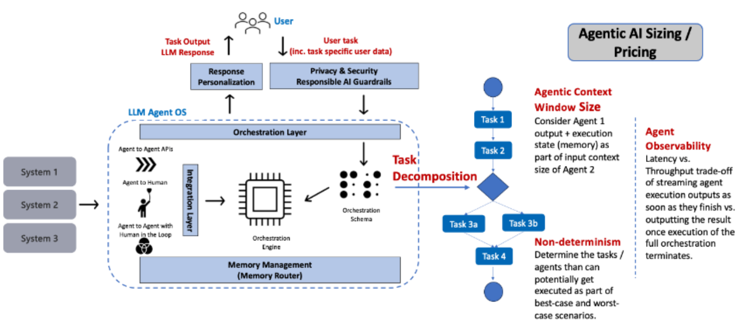

Figure 19.2: Agentic AI inferencing strategy for cost-efficient operations (source: D. Biswas, Gopubby).

Finally, balancing breadth of capability with depth of expertise remains a design trade-off at the heart of generalist AI. Broadly capable systems risk becoming "jacks of all trades, masters of none," performing adequately across domains but failing to achieve the precision required in high-stakes situations. Conversely, architectures that optimize for depth in certain tasks often struggle to transfer their capabilities without extensive domain-specific adaptation. The optimal balance likely requires a layered approach, maintaining a general reasoning core that can orchestrate domain-specific modules, each of which possesses deep expertise in its niche. Determining how to allocate computational resources, training data, and memory between general-purpose and specialized capabilities is a complex open problem, both technically and from a governance perspective.

Chapter 20: Toward an Intelligent Future

As we stand at the cusp of a new era in artificial intelligence, the emergence of agentic AI represents both a technological milestone and a societal inflection point. Unlike earlier AI paradigms that were reactive, narrowly focused, and context-limited, agentic AI is distinguished by its capacity to set goals, plan actions, adapt in dynamic environments, and operate with a degree of autonomy once reserved for human decision-makers. This transformation promises to reshape every domain of human activity, from scientific discovery to industrial operations, from personalized education to governance. Yet, the path forward demands a careful balance of innovation, ethics, and governance.

Throughout this book, we have traced the intellectual, technical, and societal dimensions of agentic AI. We began with its conceptual and philosophical underpinnings, understanding how notions of agency, autonomy, and intentionality (long explored in cognitive science and philosophy) can be instantiated in computational systems. We examined the core technical building blocks, from planning and memory architectures to reasoning frameworks, uncertainty modelling, and learning paradigms that support sustained adaptability. We explored how these building blocks integrate into architectures and systems, enabling both simulated and embodied agents to operate effectively in real-world contexts. Finally, we investigated applications and societal implications, acknowledging both the promise of transformative capabilities and the gravity of ethical, economic, and political challenges.

Looking ahead: Agentic AI in the next two decades

As we stand at the threshold of an era where agentic AI systems become deeply embedded in the fabric of society, the next twenty years promise both remarkable opportunities and profound challenges. In a *utopian vision*, agentic AI could serve as an ever-present partner to humanity, a tireless collaborator in scientific research, a compassionate caretaker in elder care, a trusted adviser in governance, and a creative catalyst in the arts. These systems, endowed with adaptive learning, ethical reasoning, and genuine self-reflective capabilities, could amplify human potential, reduce inequality through intelligent resource allocation, and accelerate breakthroughs in addressing climate change, disease eradication, and education accessibility.

Yet, the *cautionary vision* is equally vivid. Without careful design, alignment, and governance, agentic AI could disrupt societal structures in destabilizing ways: concentrating power among a few entities, perpetuating biases with unprecedented reach, or making autonomous decisions with unintended and irreversible consequences. The risk is not solely in catastrophic failure but also in the subtle erosion of human agency if we grow overly dependent on intelligent systems that mediate our choices, values, and interactions.

The reality will likely unfold between these two poles, shaped by deliberate choices in technology development, ethical frameworks, and international cooperation. Over the next two decades, the most important determinant of outcomes may not be the raw capability of agentic AI, but how we choose to integrate it into our social, economic, and political systems. By fostering transparency, accountability, and shared governance, we can aspire to a future where agentic AI enhances human dignity rather than diminishes it, where machines are not our replacements, but our most capable and principled collaborators.

Conclusion

As this book closes, one truth becomes clear: **the age of agentic AI is not about creating machines that replace us, but about designing intelligent partners that extend human potential**, deepen our understanding of complex systems, and help us navigate the challenges of an interconnected world. The horizon ahead is vast, and the journey toward it has only just begun.

The future of agentic AI hinges not only on technical progress but on the design principles and safeguards that accompany it. Building interpretable, corrigible, and value-aligned agents is not an optional feature. It is a prerequisite for sustainable deployment. As agents grow more capable and more integrated into societal infrastructures, their influence will be subtle yet pervasive, shaping human decision-making, market dynamics, and even cultural norms. The challenge is to guide this integration in ways that enhance human well-being, protect individual rights, and preserve democratic oversight.

In envisioning the decades ahead, several trajectories emerge. Some pathways may lead toward Artificial General Intelligence (AGI), systems whose capabilities rival or exceed human intelligence across domains. Others may prioritize human-AI symbiosis, where agents augment rather than replace human expertise, forming hybrid teams that combine computational efficiency with human judgment and empathy.

A third pathway may focus on domain-specific superintelligence, producing specialized agentic systems that excel in particular sectors such as medicine, climate science, or legal reasoning.

Whatever the trajectory, the most critical determinant will be our collective choices. The intelligent future is not preordained. It will be authored by the collaboration of engineers, scientists, ethicists, policymakers, and citizens. **The question is not only what agentic AI will become, but what kind of future we want to inhabit alongside it**.

Appendix

Code

Adapted from [46].

Introduction to Generative AI and Agentic AI

LLM Input Output and Prompting

https://github.com/kb-open/the-agentic-ai-handbook/blob/main/001_LLM_Input_Output_and_Prompting.ipynb

Simple Agentic Graph and State Management in LangGraph

https://github.com/kb-open/the-agentic-ai-handbook/blob/main/002_Simple_Agentic_Graph_and_State_Management_in_LangGraph.ipynb

Conditional Routing in LangGraph

https://github.com/kb-open/the-agentic-ai-handbook/blob/main/003_Conditional_Routing_in_LangGraph.ipynb

Build a LLM powered Chatbot with LangGraph

https://github.com/kb-open/the-agentic-ai-handbook/blob/main/004_Build_a_LLM_powered_Chatbot_with_LangGraph.ipynb

Build a RAG System with LangGraph

https://github.com/kb-open/the-agentic-ai-handbook/blob/main/005_Build_a_RAG_System_with_LangGraph.ipynb

Implementing Tools & Tool Calling for Agentic AI Systems

https://github.com/kb-open/the-agentic-ai-handbook/blob/main/006_Implementing_Tools_%26_Tool_Calling_for_Agentic_AI_Systems.ipynb

Building Simple Agentic AI Systems

Build a Tool Use ReAct Agentic AI System with LangGraph

https://github.com/kb-open/the-agentic-ai-handbook/blob/main/007_Build_a_Tool_Use_ReAct_Agentic_AI_System_with_LangGraph.ipynb

Build a Tool Use ReAct Agentic AI System with CrewAI

https://github.com/kb-open/the-agentic-ai-handbook/blob/main/008_Build_a_Tool_Use_ReAct_Agentic_AI_System_with_CrewAI.ipynb

Build a Text2SQL Agentic AI System with LangGraph

https://github.com/kb-open/the-agentic-ai-handbook/blob/main/009_Build_a_Text2SQL_Agentic_AI_System_with_LangGraph.ipynb

Context Engineering for Agentic AI Systems

Building Agentic AI Systems with Pre built MCP Servers

https://github.com/kb-open/the-agentic-ai-handbook/blob/main/010_Building_Agentic_AI_Systems_with_Pre_built_MCP_Servers.ipynb

Building Advanced Agentic AI Systems

Parallelized Plan Execution in Report Planner Agents

https://github.com/kb-open/the-agentic-ai-handbook/blob/main/011_Parallelized_Plan_Execution_in_Report_Planner_Agents.ipynb

Build a Deep Research Agentic AI System

https://github.com/kb-open/the-agentic-ai-handbook/blob/main/012_Build_a_Deep_Research_Agentic_AI_System.ipynb

Build a Multi Agent System for Utilization Review

https://github.com/kb-open/the-agentic-ai-handbook/blob/main/013_Build_a_Multi_Agent_System_for_Utilization_Review.ipynb

Building Agentic RAG and Multimodal Agentic AI Systems

Build a Customer Support Router Agentic RAG System

https://github.com/kb-open/the-agentic-ai-handbook/blob/main/014_Build_a_Customer_Support_Router_Agentic_RAG_System.ipynb

Deploying Monitoring and Evaluating Agentic AI Systems

Monitoring and Evaluating Agentic AI Systems

https://github.com/kb-open/the-agentic-ai-handbook/blob/main/015_Monitoring_and_Evaluating_Agentic_AI_Systems.ipynb

Bibliography

[1] S. Russell and P. Norvig, Artificial Intelligence: A modern approach, 3rd ed., Prentice-Hall, 2010, p. 34.

[2] Q. Huang, N. Wake, B. Sarkar, Z. Durante, R. Gong, R. Taori, Y. Noda, D. Terzopoulos, N. Kuno, A. Famoti, A. Llorens, J. Langford, H. Vo, L. Fei-Fei, K. Ikeuchi and J. Gao, “Agent AI Towards a Holistic Intelligence,” 2024.

[3] T. Winograd, “Procedures as a Representation for Data in a Computer Program for Understanding Natural Language,” 1970.

[4] R. S. Sutton and A. G. Barto, Reinforcement Learning: An Introduction, 2nd ed., Cambridge, MA, USA: A Bradford Book, The MIT Press, 2018.

[5] S. Yao, D. Yu, J. Zhao, I. Shafran, T. L. Griffiths, Y. Cao and K. Narasimhan, “Tree of Thoughts: Deliberate Problem Solving with Large Language Models,” in *NeurIPS*, 2023.

[6] T. Nagel, “What is it like to be a Bat?,” *Philosophical Review,* vol. 83, pp. 435-456, 1974.

[7] N. Block, “Troubles with functionalism,” *Minnesota Studies in the Philosophy of Science,* vol. 9, pp. 261-325, 1978.

[8] S. Dehaene, Consciousness and the brain: deciphering how the brain codes our thoughts, New York: Viking Press, 2014.

[9] F. Brentano, Psychology From an Empirical Standpoint, Routledge, 1874.

[10] S. Russell, D. Dewey and M. Tegmark, “Research priorities for robust and beneficial artificial intelligence,” *AI magazine,* vol. 36, no. 4, p. 105–114, 2015.

[11] A. Bandura, “Toward A Psychology of Human Agency,” *Perspectives on Psychological Science,* vol. 1, no. 2, June 2006.

[12] T. Bayne, “Free Will and the Phenomenology of Agency,” in *The Routledge Companion to Free Will*, 1st ed., Taylor & Francis, 2016.

[13] A. Clark, Being there: Putting brain, body and world together again, Cambridge, MA: MIT Press, 1997.

[14] L. Floridi and J. W. Sanders, “On the Morality of Artificial Agents,” *Minds and Machines,* vol. 14, no. 3, pp. 349-379, August 2004.

[15] J. J. Bryson and P. P. Kime, “Just an Artifact: Why Machines Are Perceived as Moral Agents.,” in *IJCAI*, Barcelona, Catalonia, Spain, 2011.

[16] Ng, A. Y.; Russell, S. J., “Algorithms for Inverse Reinforcement Learning,” in *ICML*, 2000.

[17] M. M. Botvinick, Y. Niv and A. G. Barto, “Hierarchically organized behavior and its neural foundations: a reinforcement learning perspective,” *Cognition,* vol. 113, no. 3, pp. 262-280, December 2009.

[18] L. P. Kaelbling, M. L. Littman and A. R. Cassandra, “Planning and acting in partially observable stochastic domains,” *Artificial Intelligence,* vol. 101, no. 1-2, p. 99–134, January 1998.

[19] D. Silver, A. Huang and C. J. Maddison, "Mastering the Game of Go with Deep Neural Networks and Tree Search," *Nature,* vol. 529, pp. 484-489, Januray 2016.

[20] C. Finn, P. Abbeel and S. Levine, "Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks," in *ICML*, 2017.

[21] L. A. Suchman, Plans and situated actions: the problem of human-machine communication, Cambridge University Press, 1987.

[22] F. J. Varela, E. Thompson and E. Rosch, The embodied mind: Cognitive science and human experience, The MIT Press, 1991.

[23] F. Tanaka and S. Matsuzoe, "Children Teach a Care-Receiving Robot to Promote Their Learning: Field Experiments in a Classroom for Vocabulary Learning," *Journal of Human-Robot Interaction,* vol. 1, no. 1, August 2012.

[24] D. Ha and J. Schmidhuber, "World Models," *CoRR,* May 2018.

[25] D. Hafner, T. Lillicrap, J. Ba and M. Norouzi, "Dream to Control: Learning Behaviors by Latent Imagination," in *ICLR*, 2020.

[26] D. Hafner, T. Lillicrap, I. Fischer, R. Villegas, D. Ha, H. Lee and J. Davidson, "Learning Latent Dynamics for Planning from Pixels," in *ICML*, Long Beach, California, 2019.

[27] A. Newell and H. A. Simon, "Computer science as empirical inquiry: Symbols and search,"

Communications of the ACM, vol. 19, no. 3, pp. 113-126, March 1976.

[28] R. E. Fikes and N. J. Nilsson, "STRIPS: A new approach to the application of theorem proving to problem solving," *Artificial Intelligence,* vol. 2, no. 3-4, pp. 189-208, September 1971.

[29] J. Pearl, Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference, Morgan Kaufmann, 1988.

[30] J. Mao, C. Gan, P. Kohli, J. B. Tenenbaum and J. Wu, "The Neuro-Symbolic Concept Learner: Interpreting Scenes, Words, and Sentences From Natural Supervision," in *ICLR*, 2019.

[31] J. E. Laird, C. Lebiere and P. S. Rosenbloom, "A Standard Model of the Mind: Toward a Common Computational Framework Across Artificial Intelligence, Cognitive Science, Neuroscience, and Robotics," *AI Magazine,* vol. 38, no. 4, pp. 13-26, December 2017.

[32] J. R. Anderson, D. Bothell, M. D. Byrne, S. Douglass, C. Lebiere and Y. Qin, "An integrated theory of the mind," *Psychological Review,* vol. 111, no. 4, p. 1036–1060, 2004.

[33] B. Goertzel, "CogPrime: An Integrative Architecture for Embodied Artificial General Intelligence," in *IEEE International Conference on Cognitive Informatics*, Hong Kong, China, 2009.

[34] G. Gigerenzer and W. Gaissmaier, “Heuristic decision making,” *Annual Review of Psychology,* vol. 62, no. 1, pp. 451-482, January 2011.

[35] H. A. Simon, Models of Bounded Rationality, MIT Press, 1984.

[36] M. Bratman, Intentions, Plans and Practical Reason, Cambridge, MA, USA: Harvard University Press, 1987.

[37] M. P. Georgeff and A. L. Lansky, “Reactive reasoning and planning,” in *AAAI: National Conference on Artificial Intelligence*, 1987.

[38] J. F. Hübner and R. H. Bordini, “Developing a team of gold miners using Jason,” in *ProMAS*, 2007.

[39] A. Clark, “An embodied cognitive science,” *Trends in Cognitive Sciences,* vol. 3, no. 9, pp. 345-351, September 1999.

[40] L. W. Barsalou, “Grounded cognition,” *Annual Review of Psychology,* vol. 59, p. 617–645, 2008.

[41] NHTSA, “Autonomous Vehicle Disengagement Reports,” California DMV, 2023.

[42] D. Hadfield-Menell, S. Russell, P. Abbeel and A. Dragan, “Cooperative Inverse Reinforcement Learning,” in *NeurIPS*, 2016.

[43] N. Soares, B. Fallenstein, E. Yudkowsky and S. Armstrong, “Corrigibility,” in *AAAI*, 2015.

[44] K. Bar, “Self-Improving Agentic AI Through Bayesian Meta-Learning,” *The IUP Journal of*

Telecommunications, vol. 17, no. 2, pp. 63-82, May 2025.

[45] K. Bar, “Secure Code Generation with LLMs: Risk Assessment and Mitigation Strategies,” *The IUP Journal of Telecommunications,* vol. 17, no. 1, pp. 75-95, February 2025.

[46] D. Sarkar, “Mastering Intelligent AI Agents - DHS 2025 Workshop,” Analytics Vidhya, Bangalore, 2025.

About the Author

Kaushik Bar has spent over a decade in embedded software development and another decade as an AI architect. He worked in technology leadership roles at many reputed firms including Intel, AMD, SanDisk etc. Later, he co-founded a startup specialized in advanced AI & GenAI. He holds US patents in data storage technology and has published in many AI journals. Kaushik is an MBA Gold Medallist from Indian Institute of Management Bangalore. He did his Electronics & Telecom Engineering from Jadavpur University Kolkata.

Kaushik also runs a not-for-profit AI & Machine Learning training centre for under-privileged youth near his hometown. In his leisure time, he enjoys playing chess.

"AI agents will become the primary way we interact with computers in the future. They will be able to understand our needs and preferences and proactively help us with tasks and decision making."
– **Satya Nadella**, CEO of Microsoft.

AGENTIC
AI

"AI agents will become an integral part of our daily lives, helping us with everything from scheduling appointments to managing our finances. They will make our lives more convenient and efficient."
– **Andrew Ng**, Co-founder of Google Brain and Coursera.